{3} Current Electricity

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* INTRODUCTION

* ELECTRIC CURRENT

* ELECTRIC CURRENTS IN CONDUCTORS

* OHM' S LAW

* DRIFT OF ELECTRONS AND THE ORIGIN OF RESISTIVITY

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* TEMPERATURE DEPENDENCE OF RESISTIVITY
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* In the present chapter, we shall study some of the basic laws concerning steady electric currents.

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* The direction of current is taken as the direction of motion +ve charges or opposite to the motion of -ve charge.

Ex 1 : In cosmic rays 0.15 protons/cm^2-s are entering the earth's atmosphere. If the radius of the earth is 6400 km, what is the current received by the earth in the form of cosmic rays?

Ex 2: In neon gas discharge tube 2.9×10^{18} Ne^{+} ions move to the right through a cross-section of the tube each second, while 1.2×10^{18} electrons move to the left in this time. What is the net electric current in the tube?

Ex 3 : In the Bohr's model of hydrogen atom, the electron is assumed to rotate in a circular orbit of radius 5×10^{-11} m, at a speed of 2.2×10^{6} m/s. What is the current associated with electron motion?

Ex 4 : The current in a wire varies with time according to the relation $I = 2\ \text{A} + (0.6\ \text{A/s}^2)\ t^2$.

(a) How many coulomb of charge pass a cross-section of the wire in the time interval between $t = 0$ and $t = 10$ s?

(b) What constant current would transpot the same charge in same charge in the same time interval?

Ex 5 : A current through a wire depends on time as $I = \alpha_0 \; t + \beta \; t^2$. Where $\alpha_0 = 20 \; A/s$ and $\beta = 8 \; A \; s^{-2}$. Find the charge crossed through a section of the wire in 15 s.

(a) 2100 C (b) 2250 C (c) 260 C (d) 1125 C

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*** In our discussions, we will focus only on solid conductors so that the current is carried by the negatively charged electrons in the background of fixed positive ions.**

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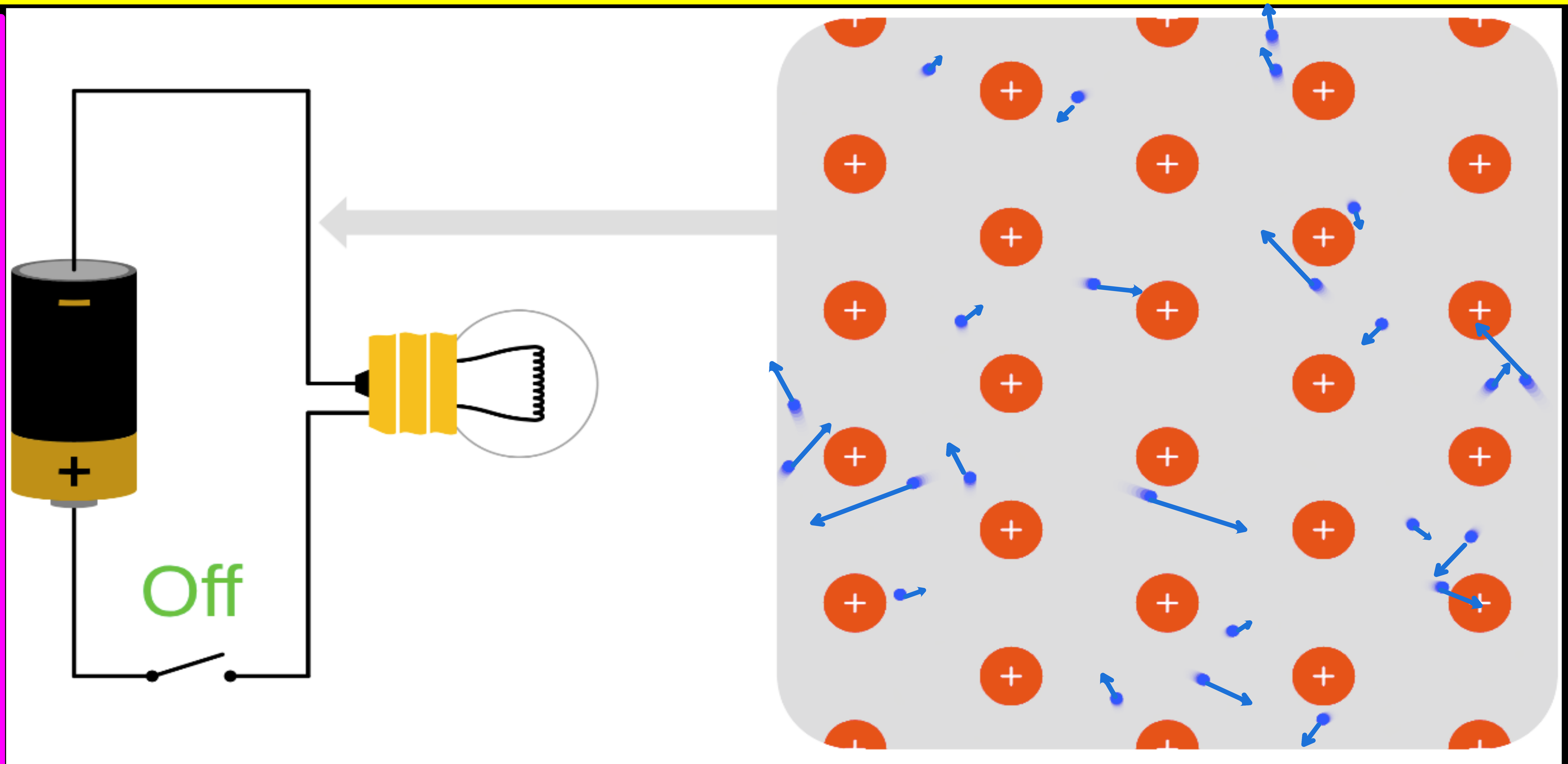

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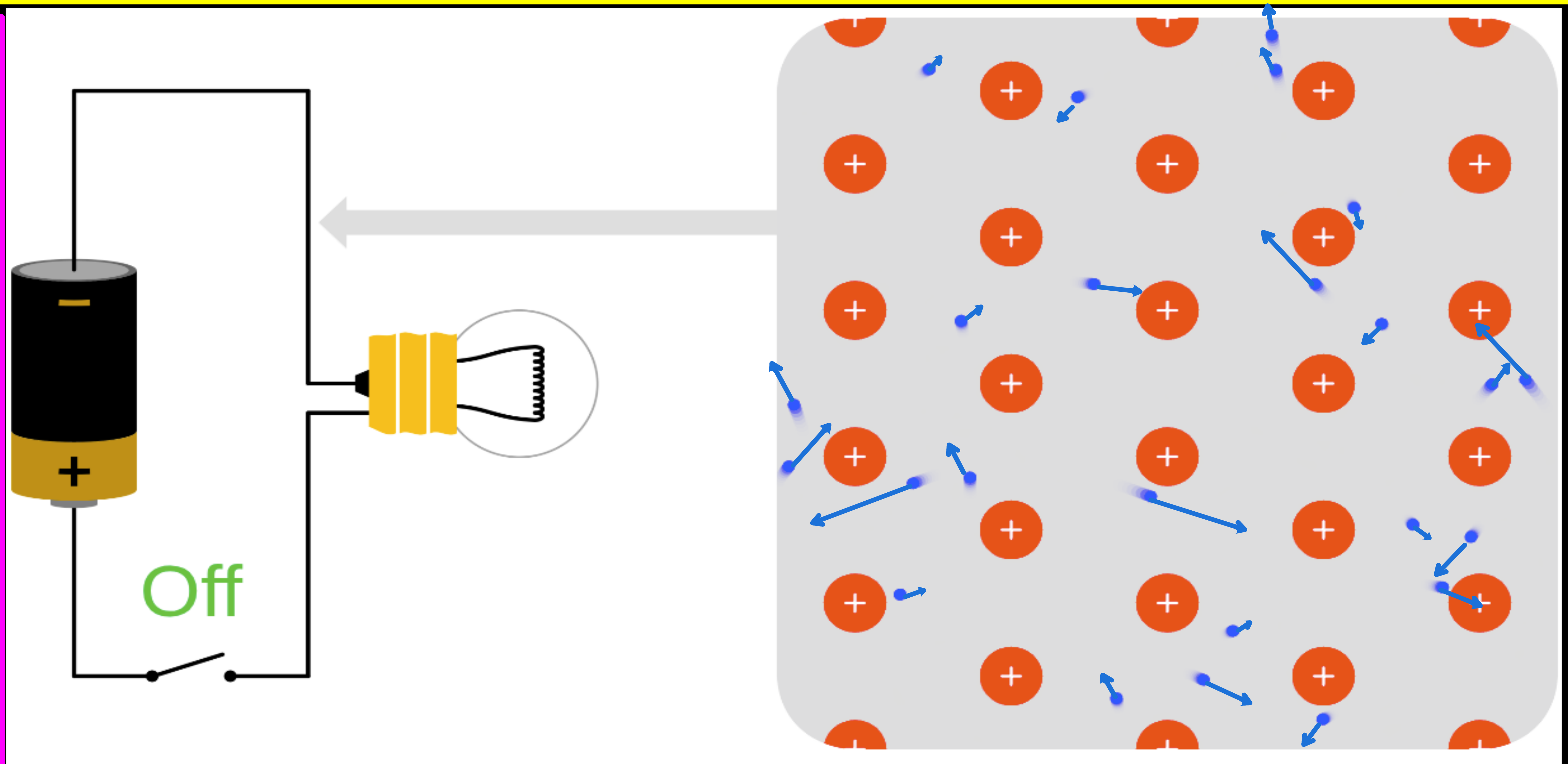

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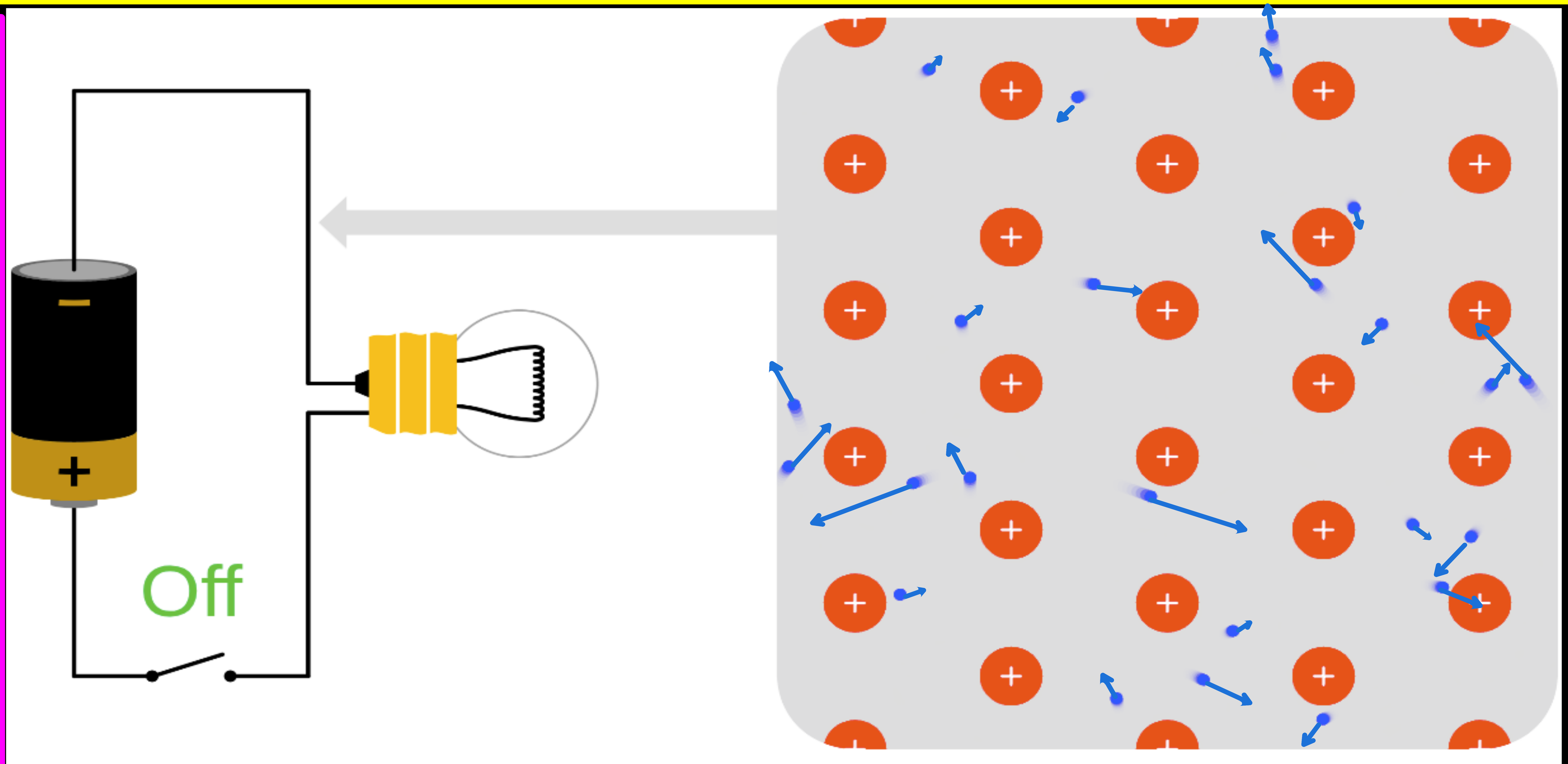

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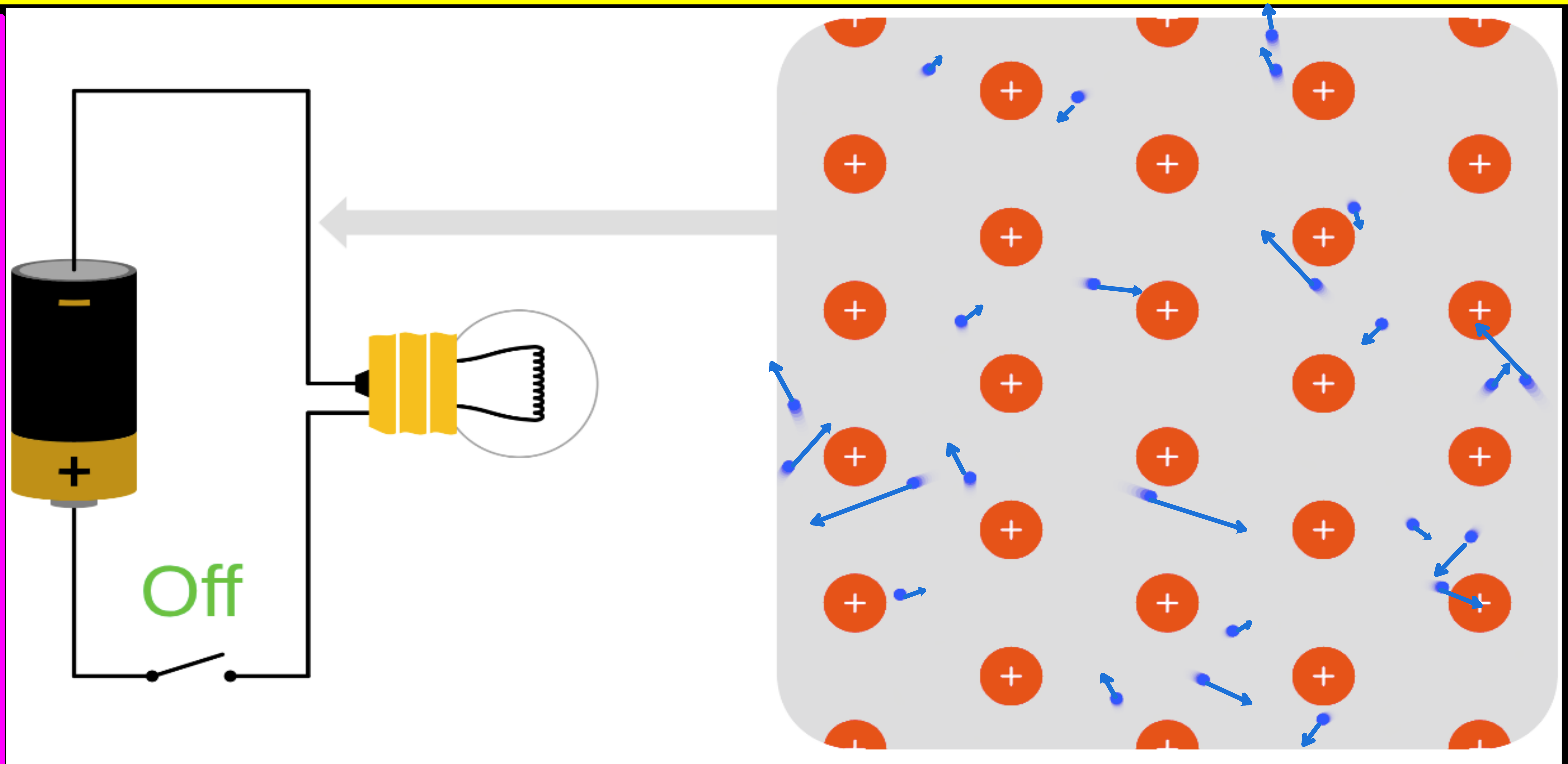

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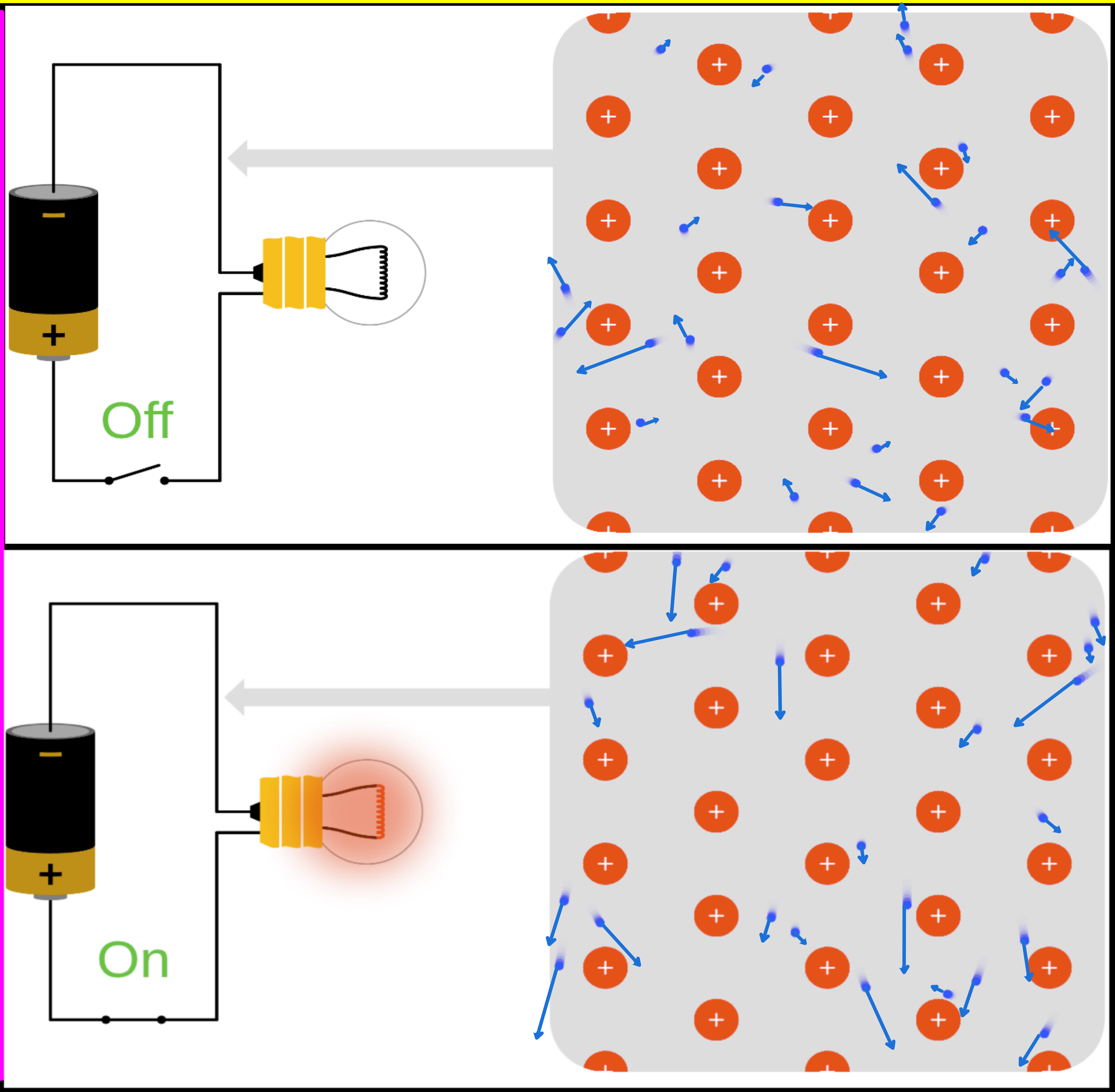
Case 2 : When elecrtirc filed is present :

* The electrons will be accelerated due to this field, in the opposite direction of the field. The electrons, as long as they are moving, will constitute an electric current.



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$$R \propto \frac{l}{A} \text{ or } R = \rho\frac{l}{A}$$

* where the constant of proportionality called resistivity (ρ) depends on the material

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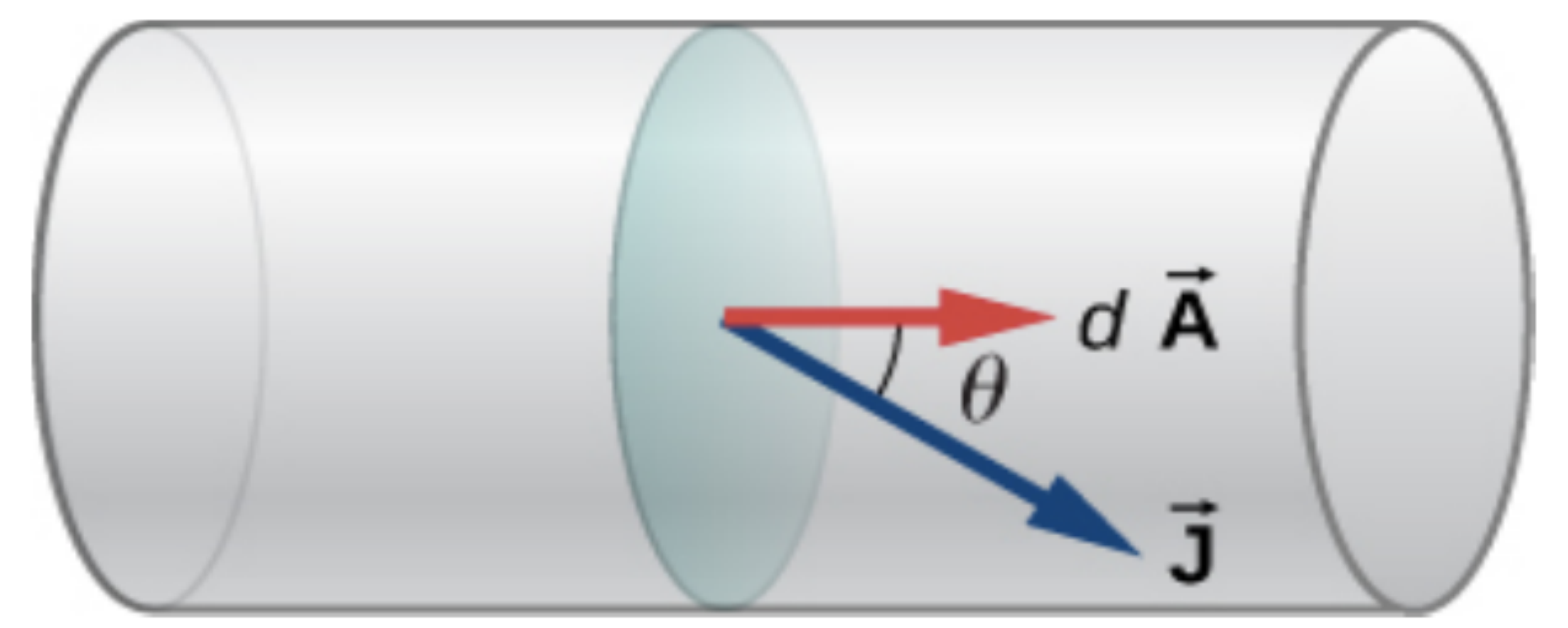

Figure 9.12 The current density $\vec{J}$ is defined as the current passing through an infinitesimal cross-sectional area divided by the area. The direction of the current density is the direction of the net flow of positive charges and the magnitude is equal to the current divided by the infinitesimal area.

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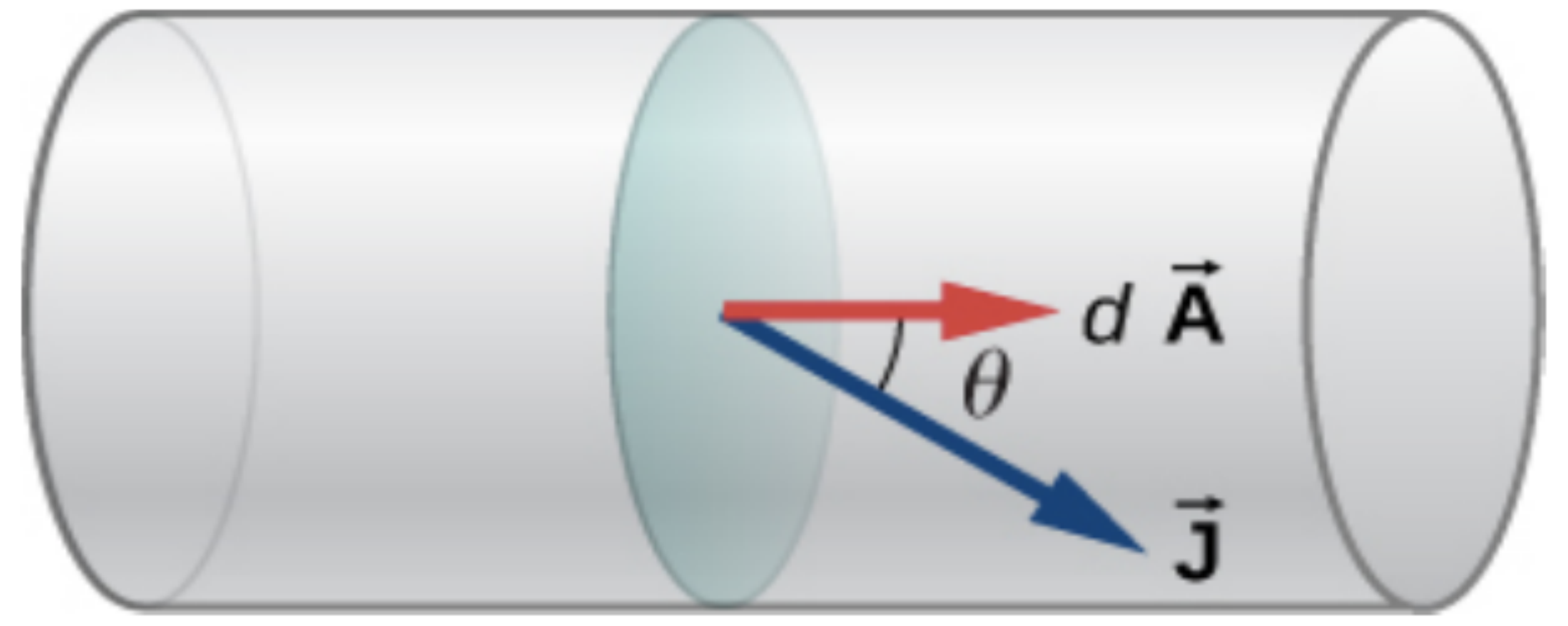

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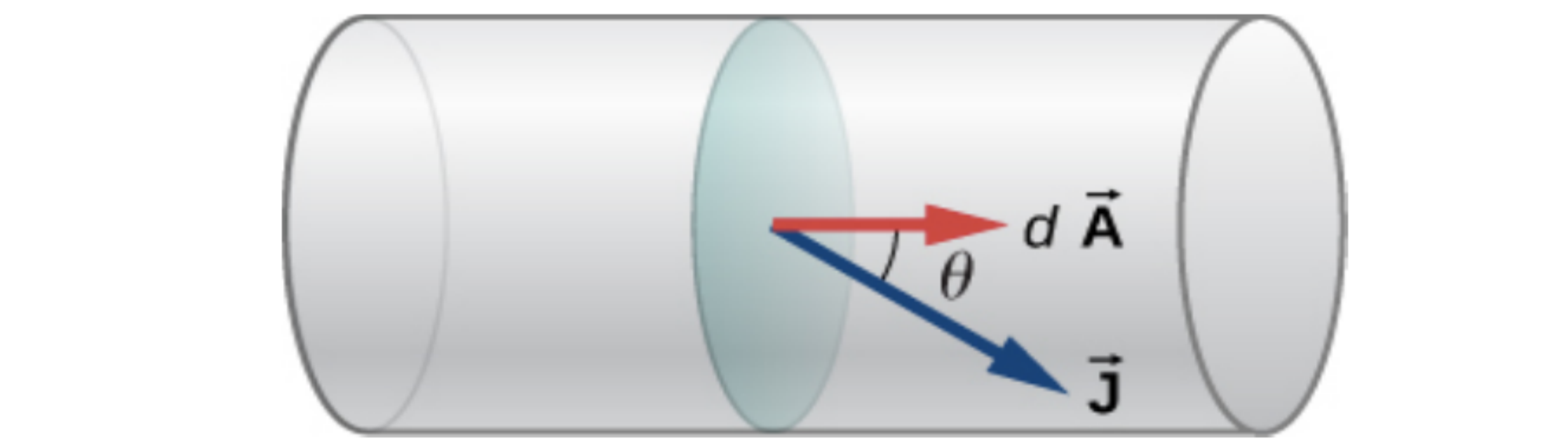

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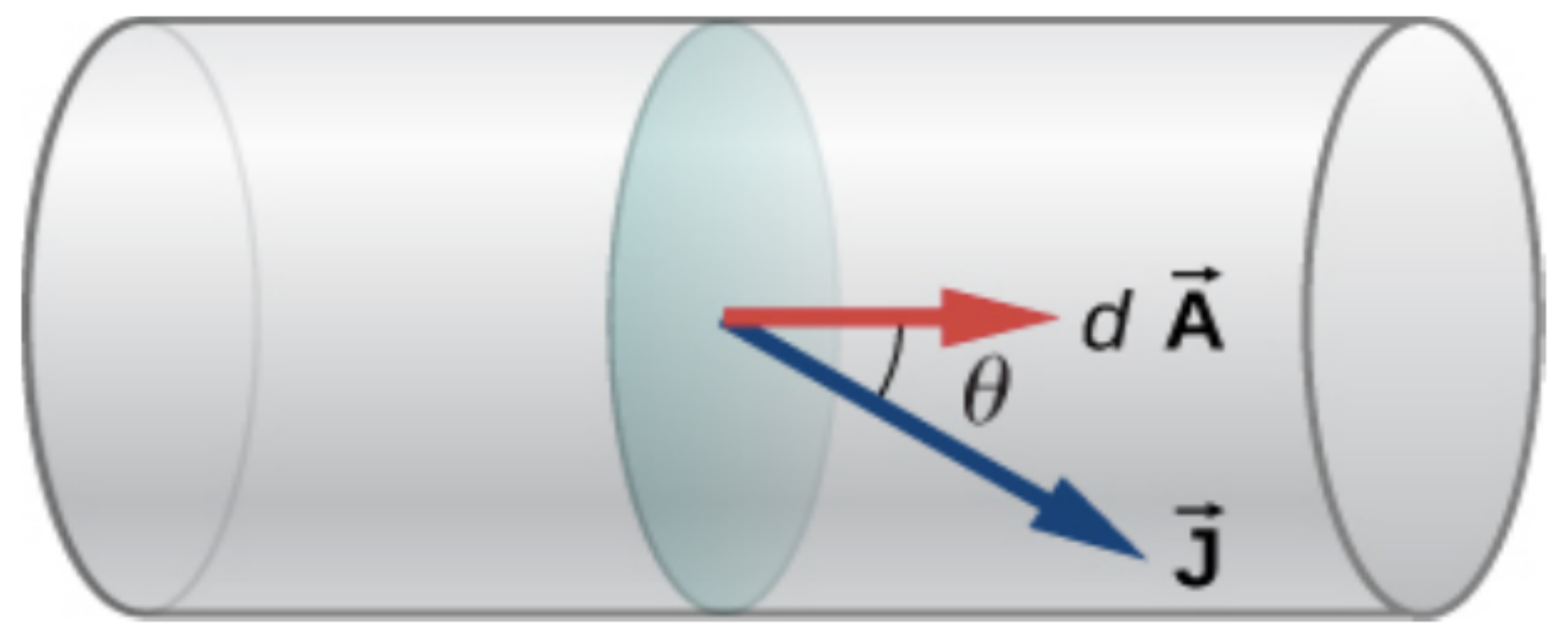

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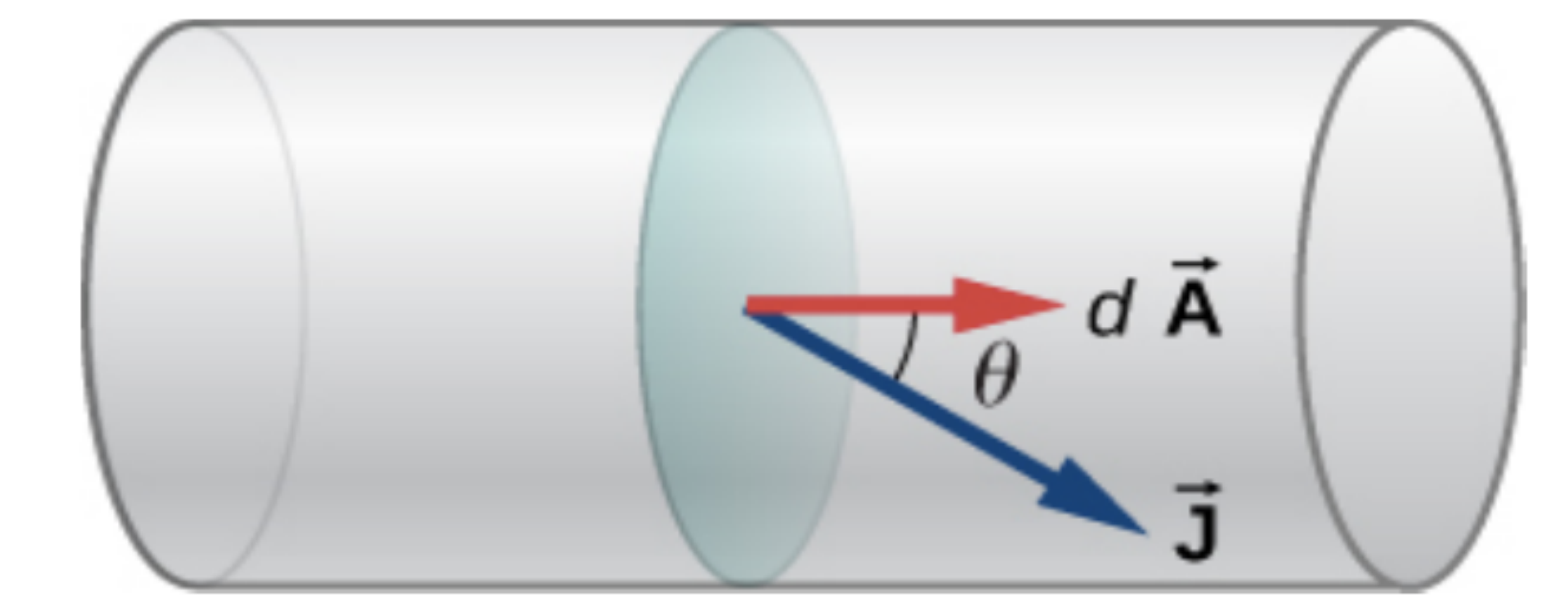

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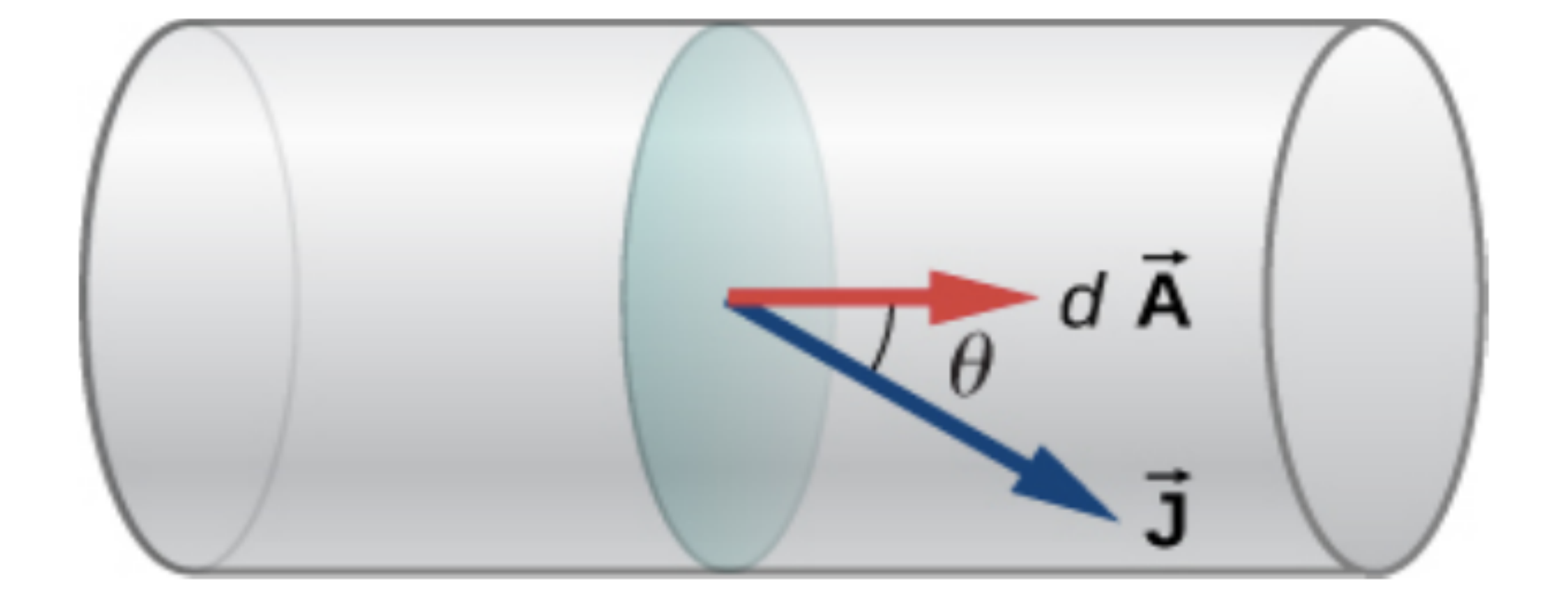

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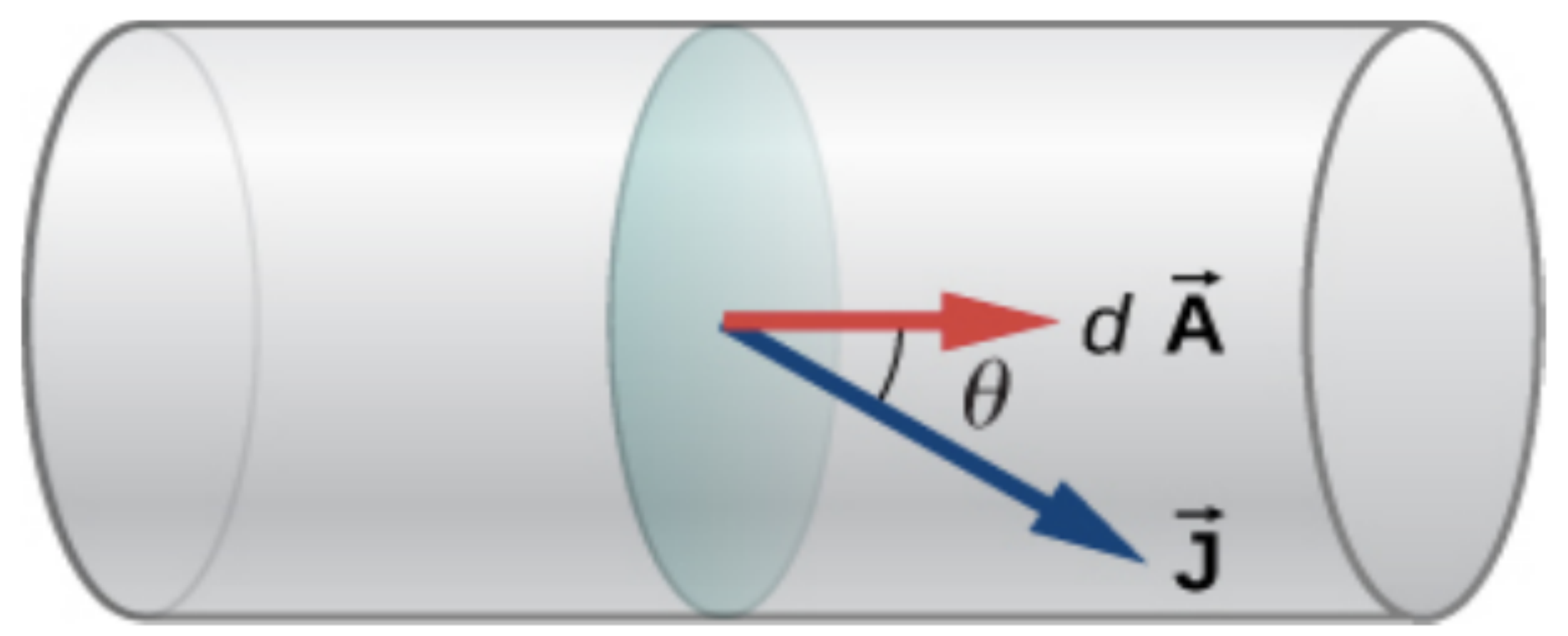

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3.6 A negligibly small current is passed through a wire of length 15 m and uniform cross-section 6.0×10^{-7} m^2, and its resistance is measured to be 5.0 Ω. What is the resistivity of the material at the temperature of the experiment?

Ex 6 : A wire 50 cm long and 1 mm² in cross-section carries a current of 4 A when connected to a 2 V battery. The resistivity of the wire is

$a)\ 2 \times 10^{-7}\ \Omega m$ $\qquad b)\ 5 \times 10^{-7}\ \Omega m$

$c)\ 4 \times 10^{-6}\ \Omega m$ $\qquad d)\ 1 \times 10^{-6}\ \Omega m$

Ex 7: A wire of length L and cross-section area A has resistance R. What will be the resistance of the wire if it is stretched to twice its original length? Assume that the density and resistivity of material do not change when the wire is stretched.

Ex 8 : A wire of length L is drawn such that its diameter is reduced to half of its original diameter. If the initial resistance of the wire is 10 Ω, its new resistance would be

a) 40 Ω (b) 80 Ω (c) 120 Ω (d) 160 Ω

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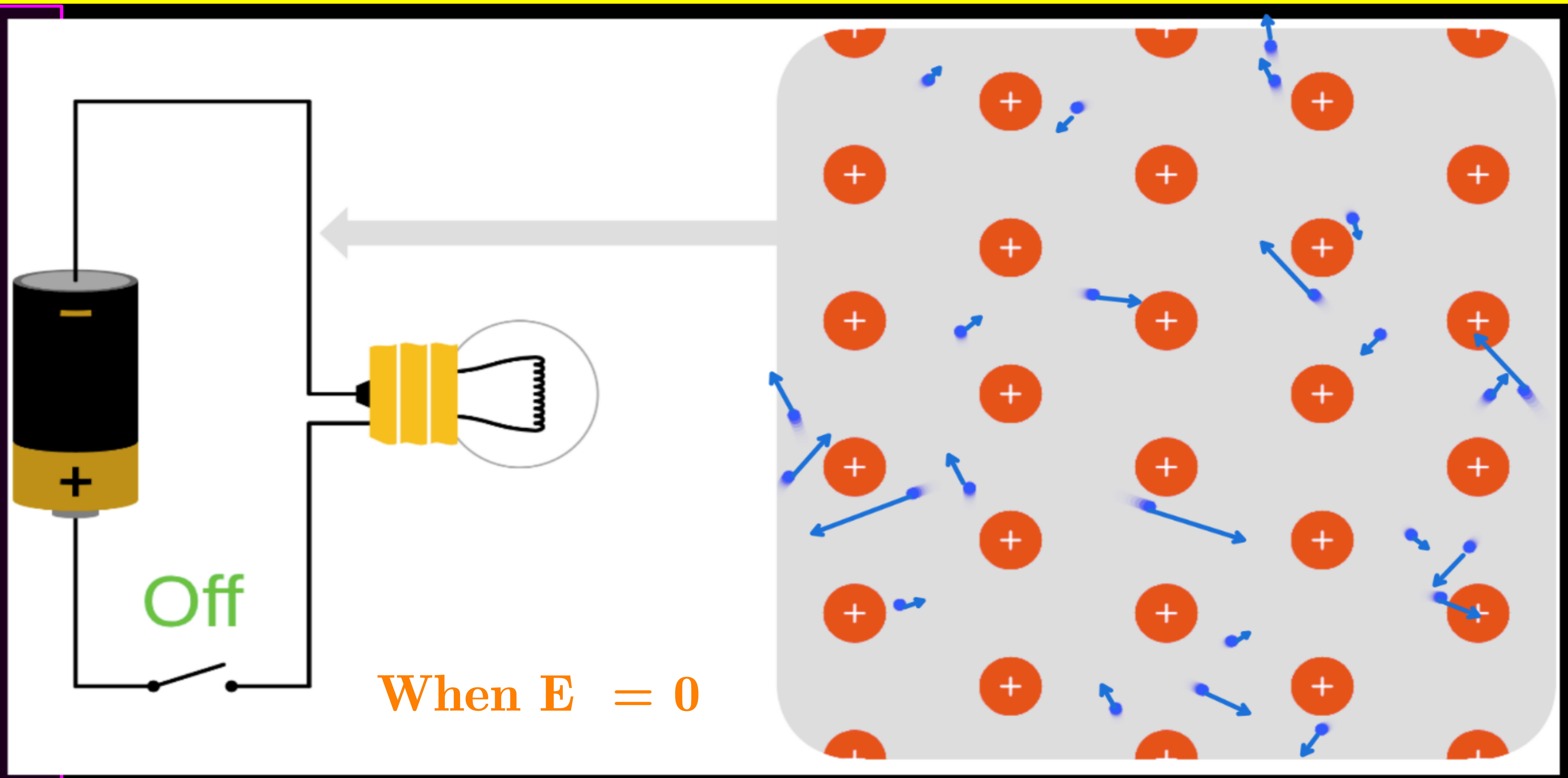

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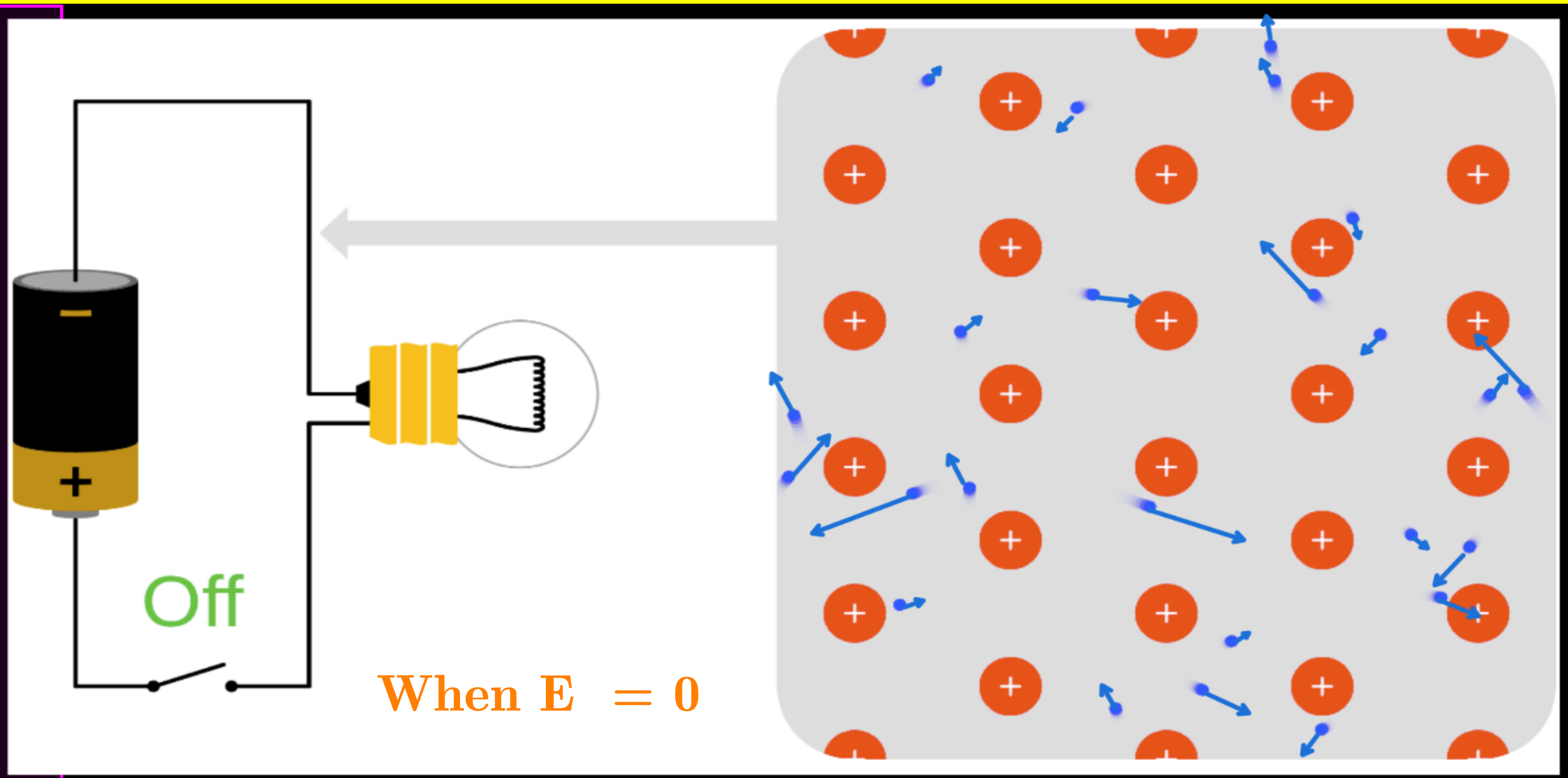

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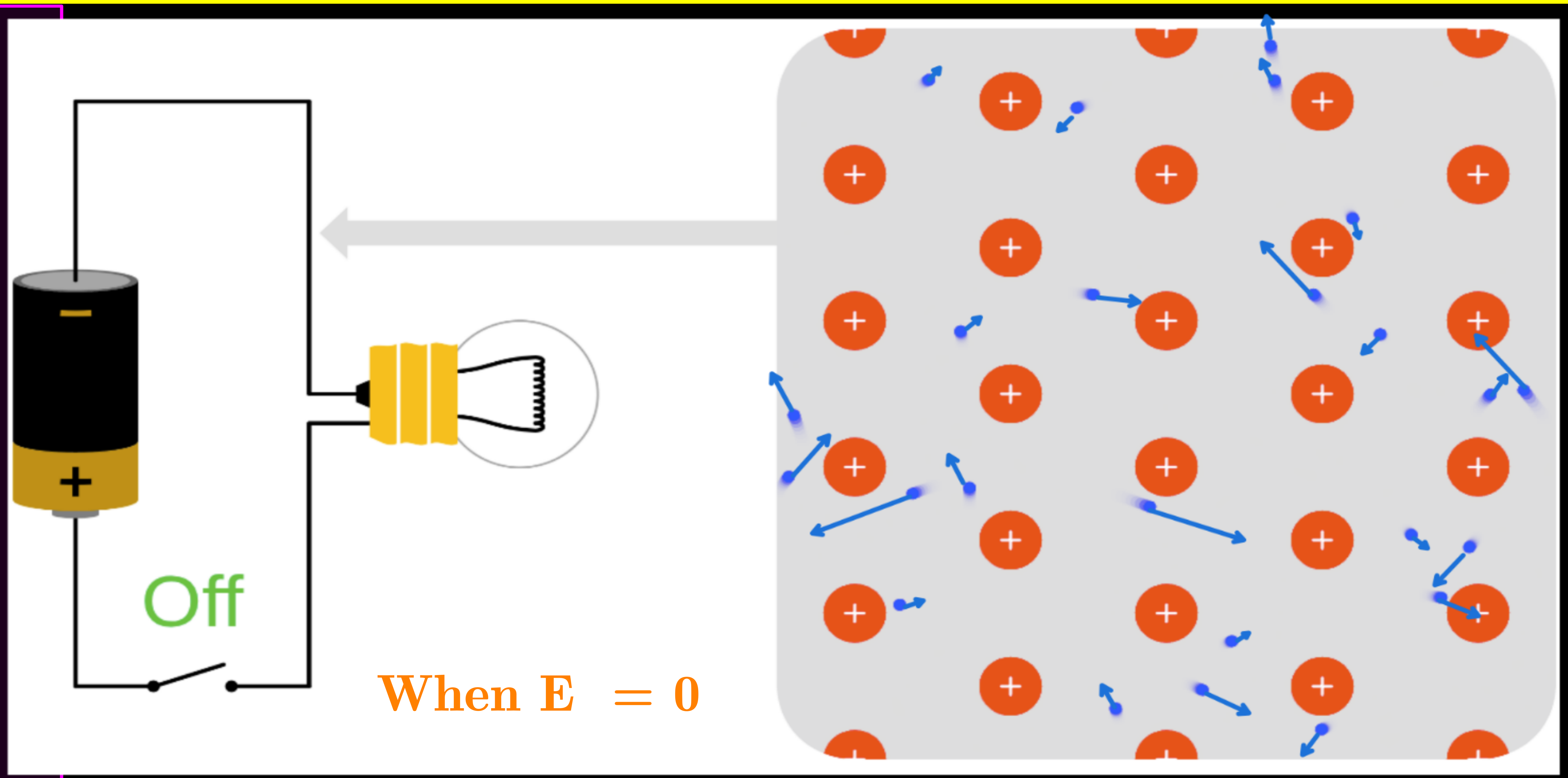
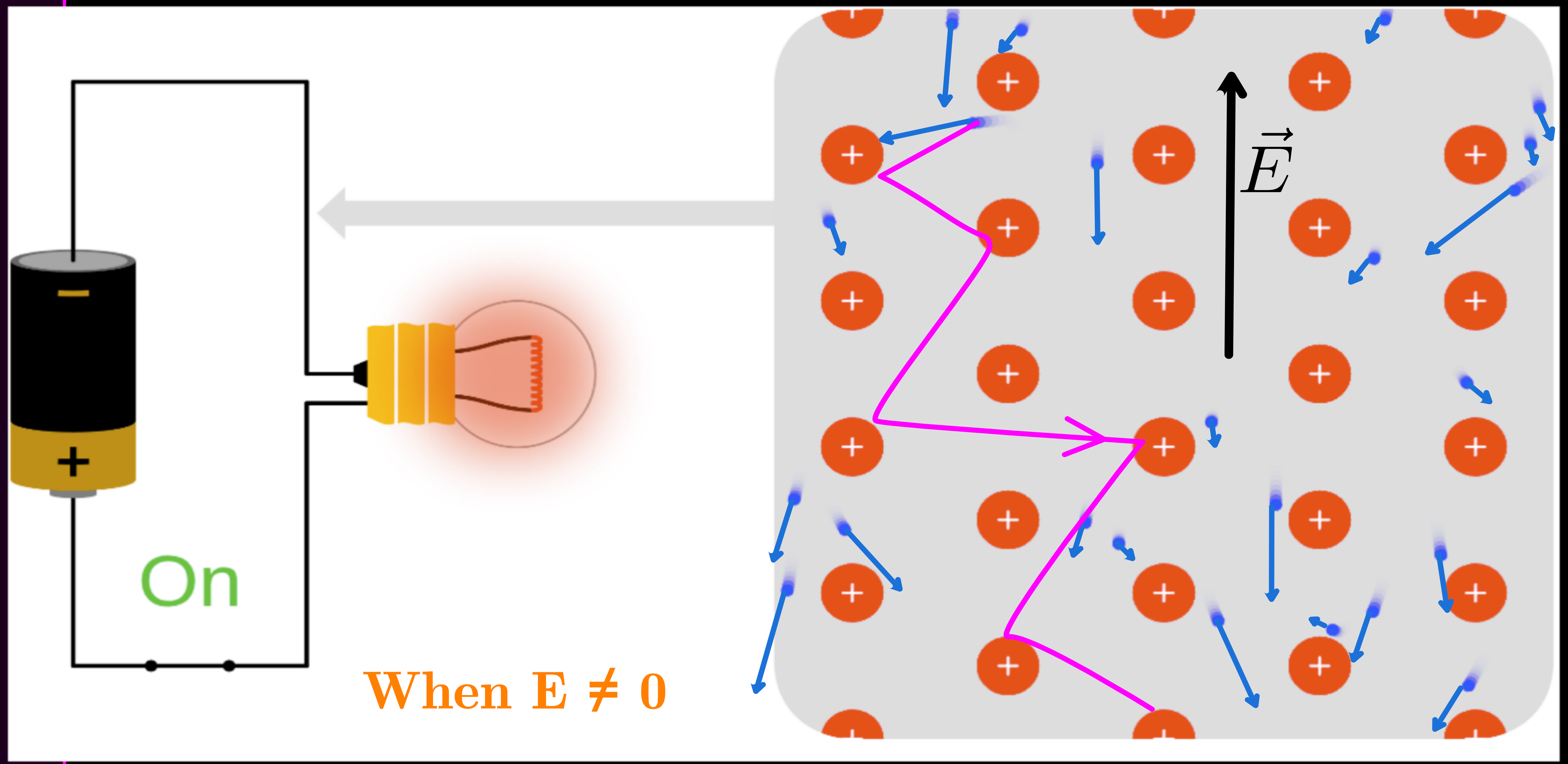
$$\vec{F} = -e\vec{E}$$





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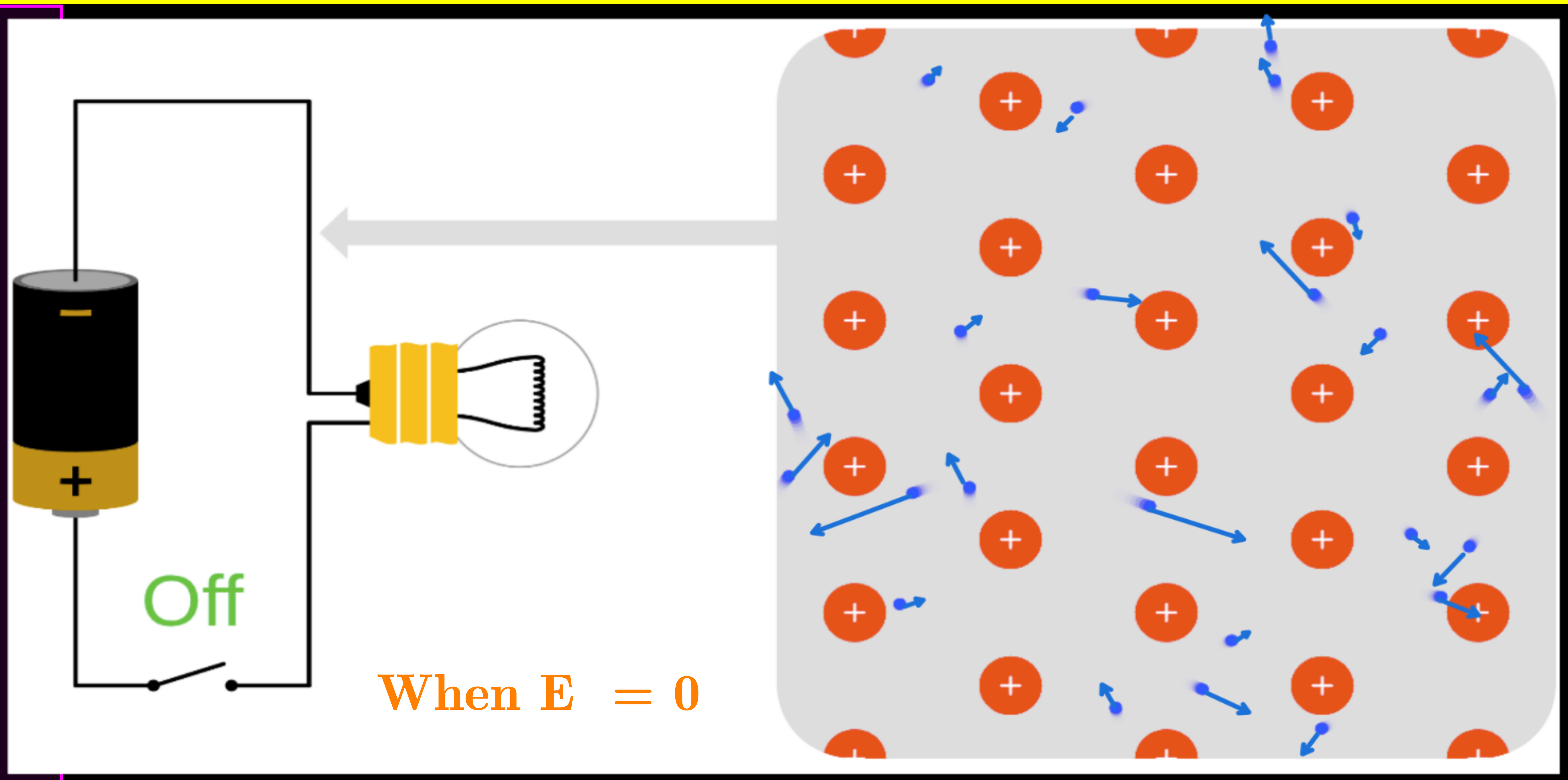

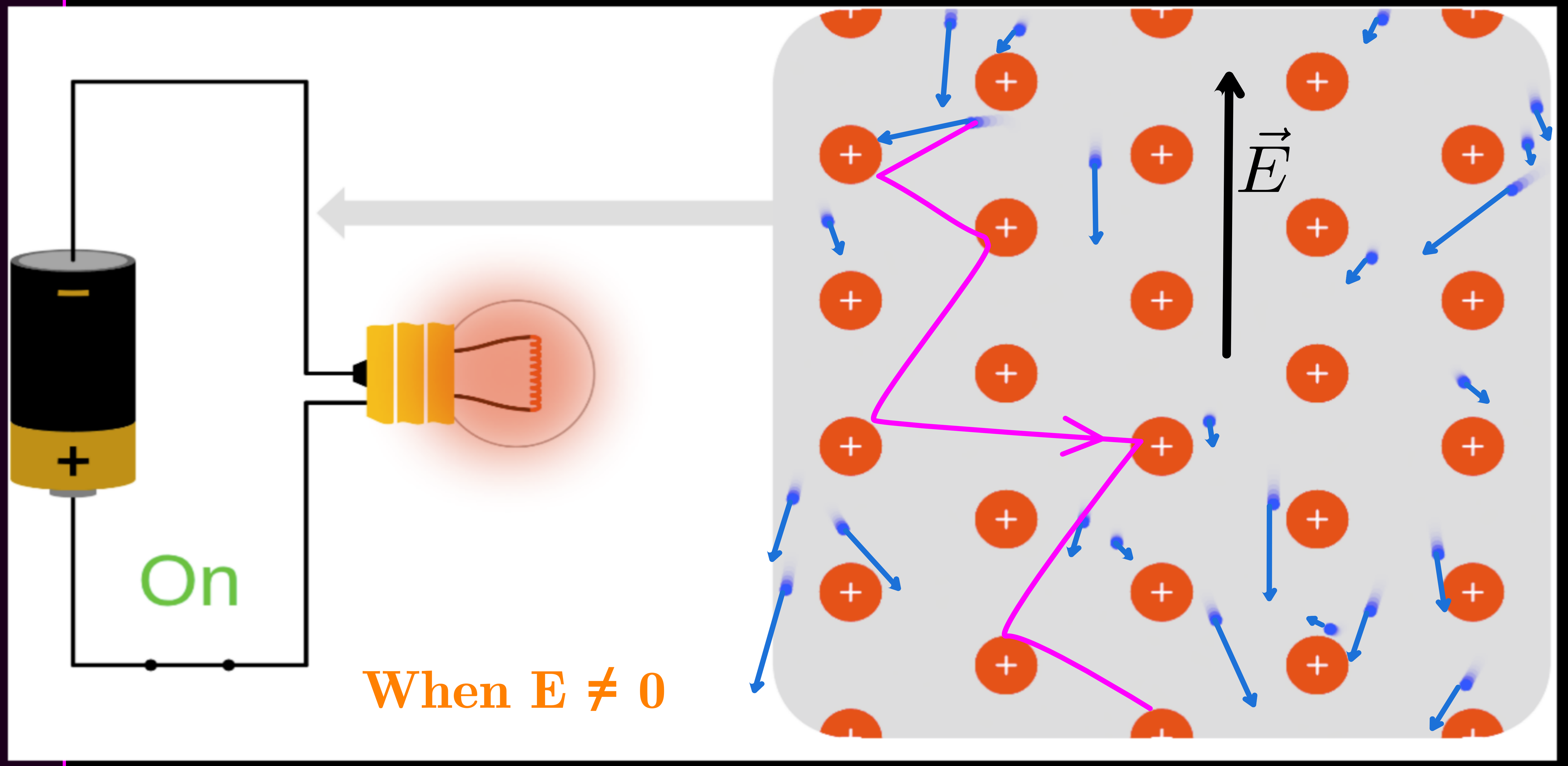

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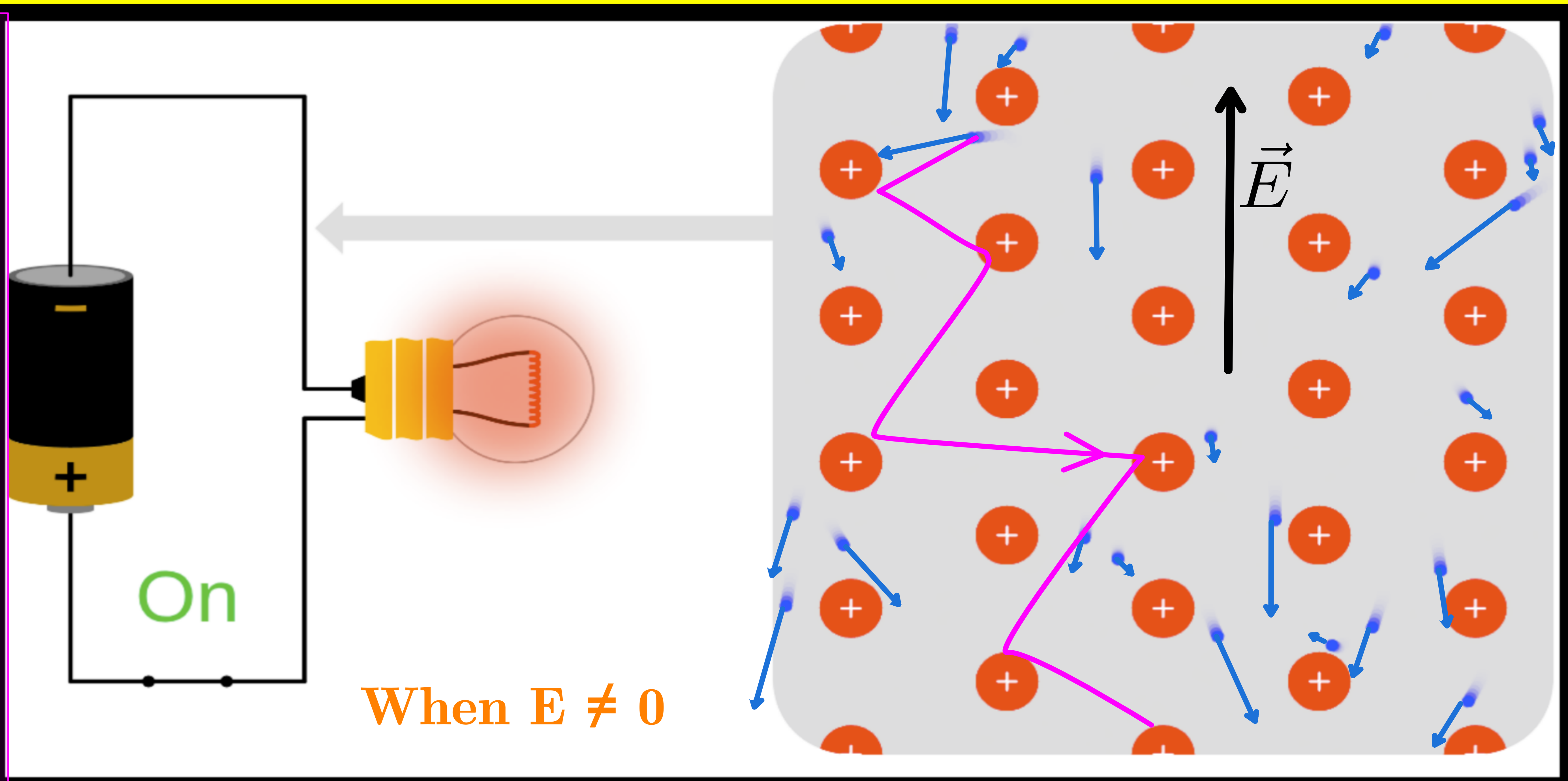

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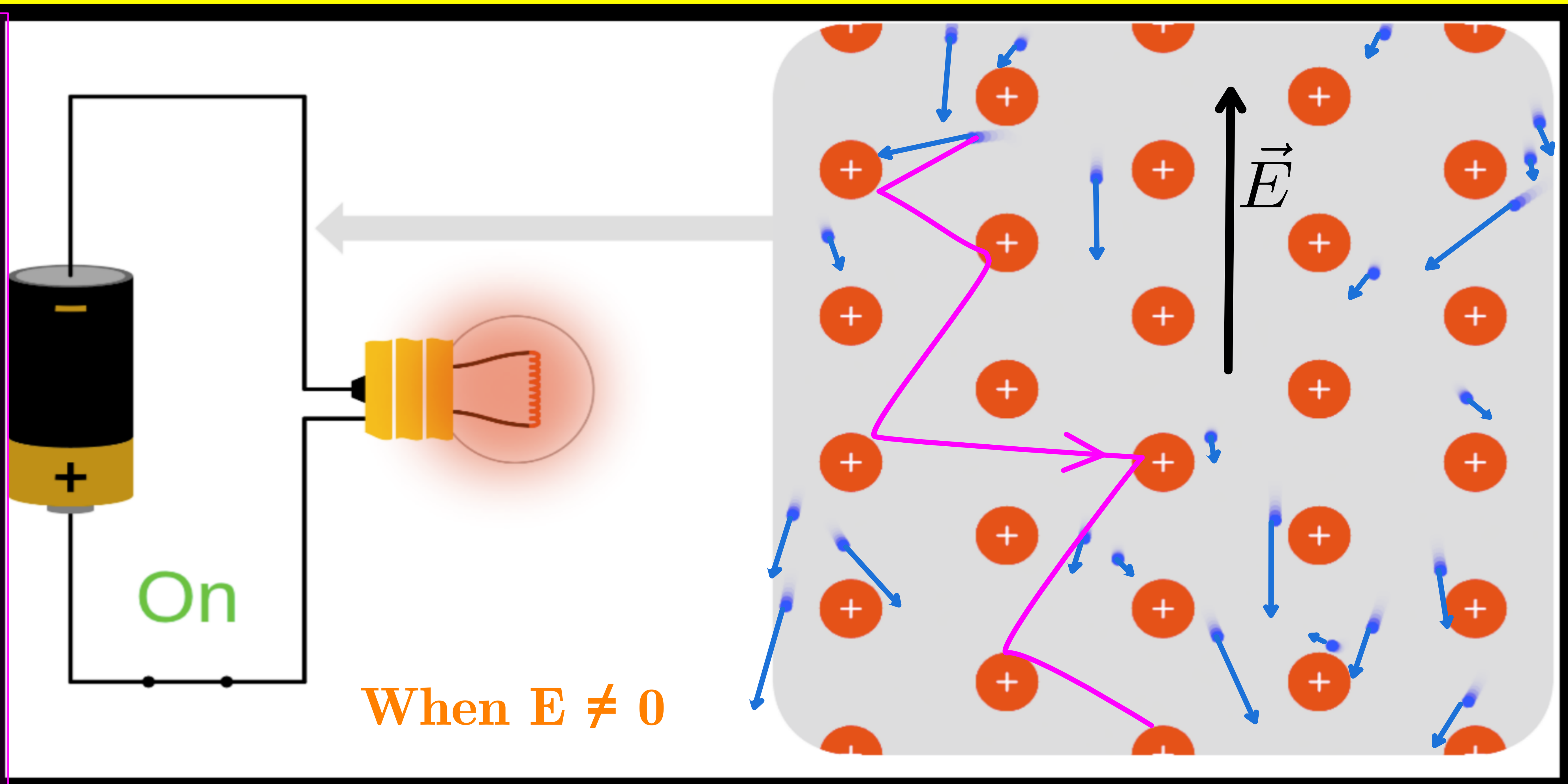
$$\vec{v}_i = \vec{u}_i + \vec{a}\ t_i$$



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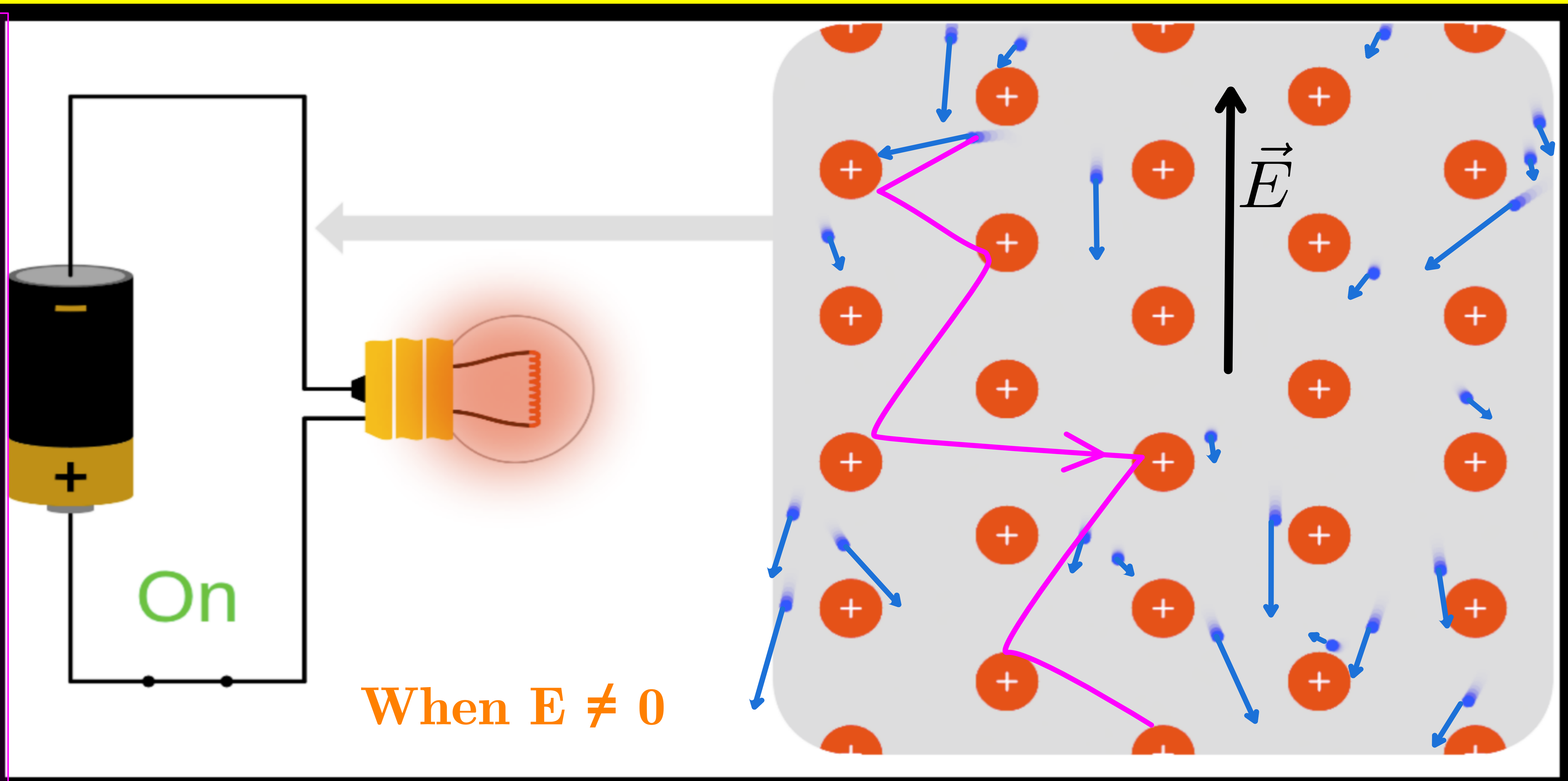

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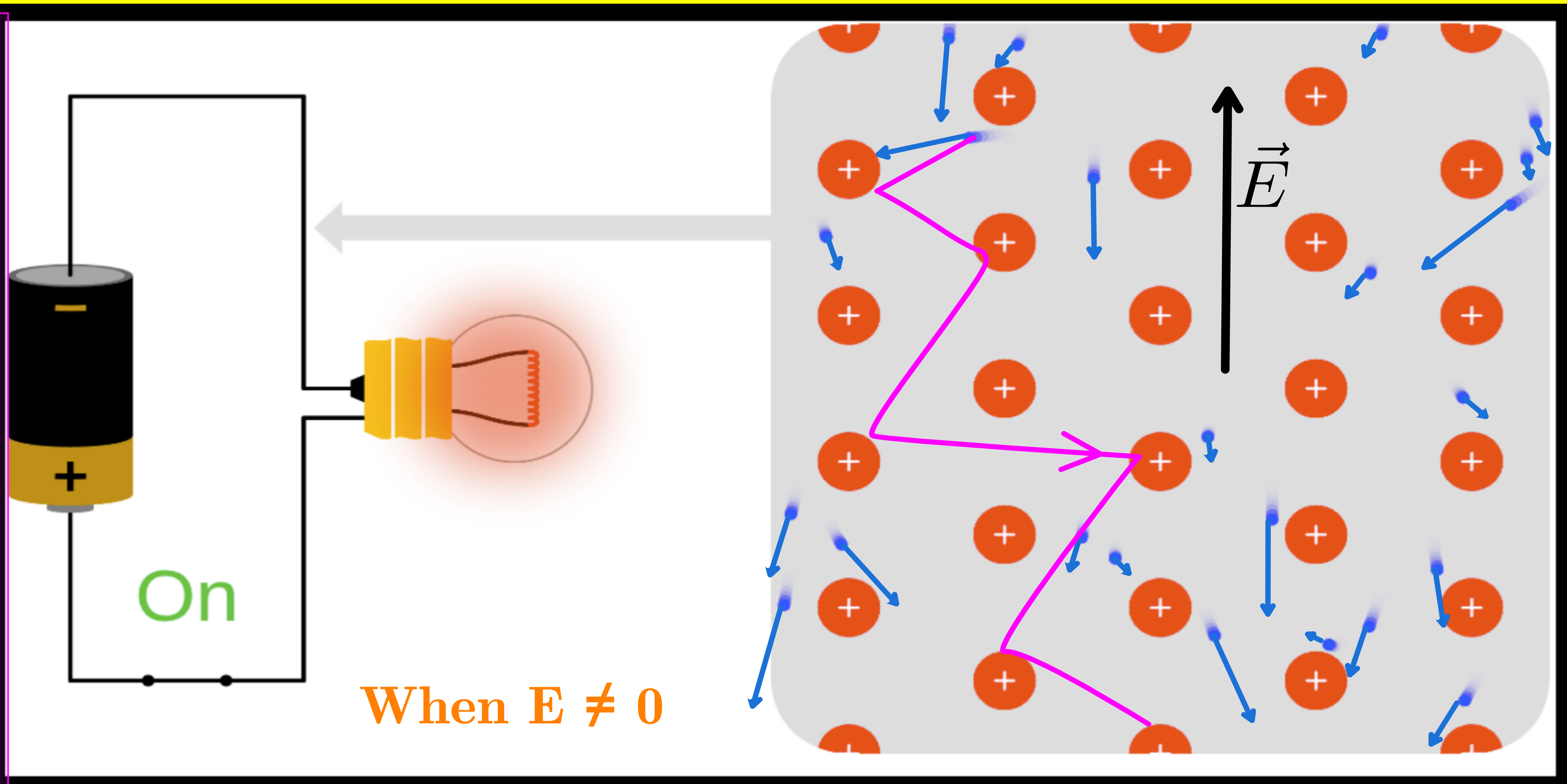

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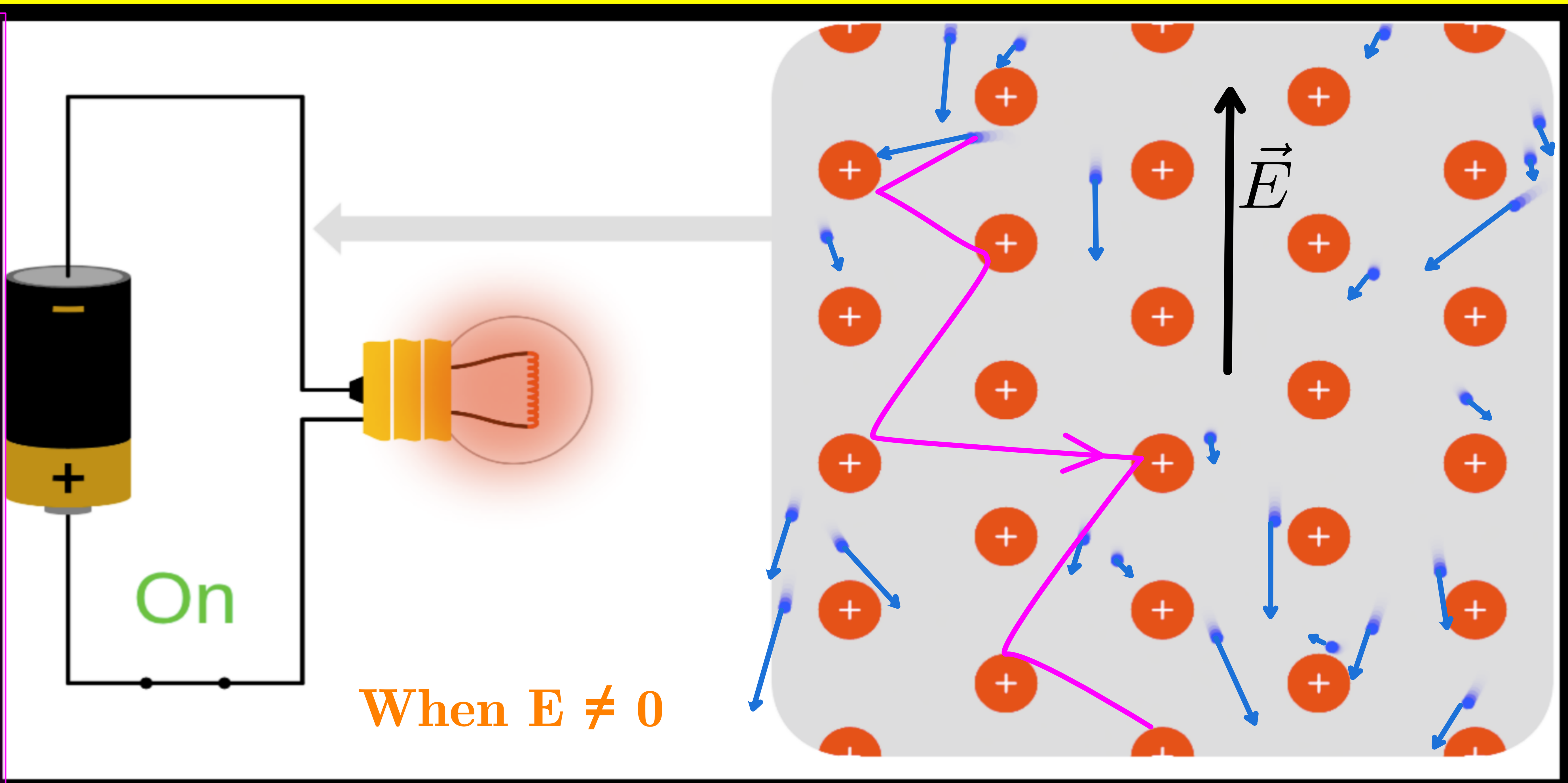

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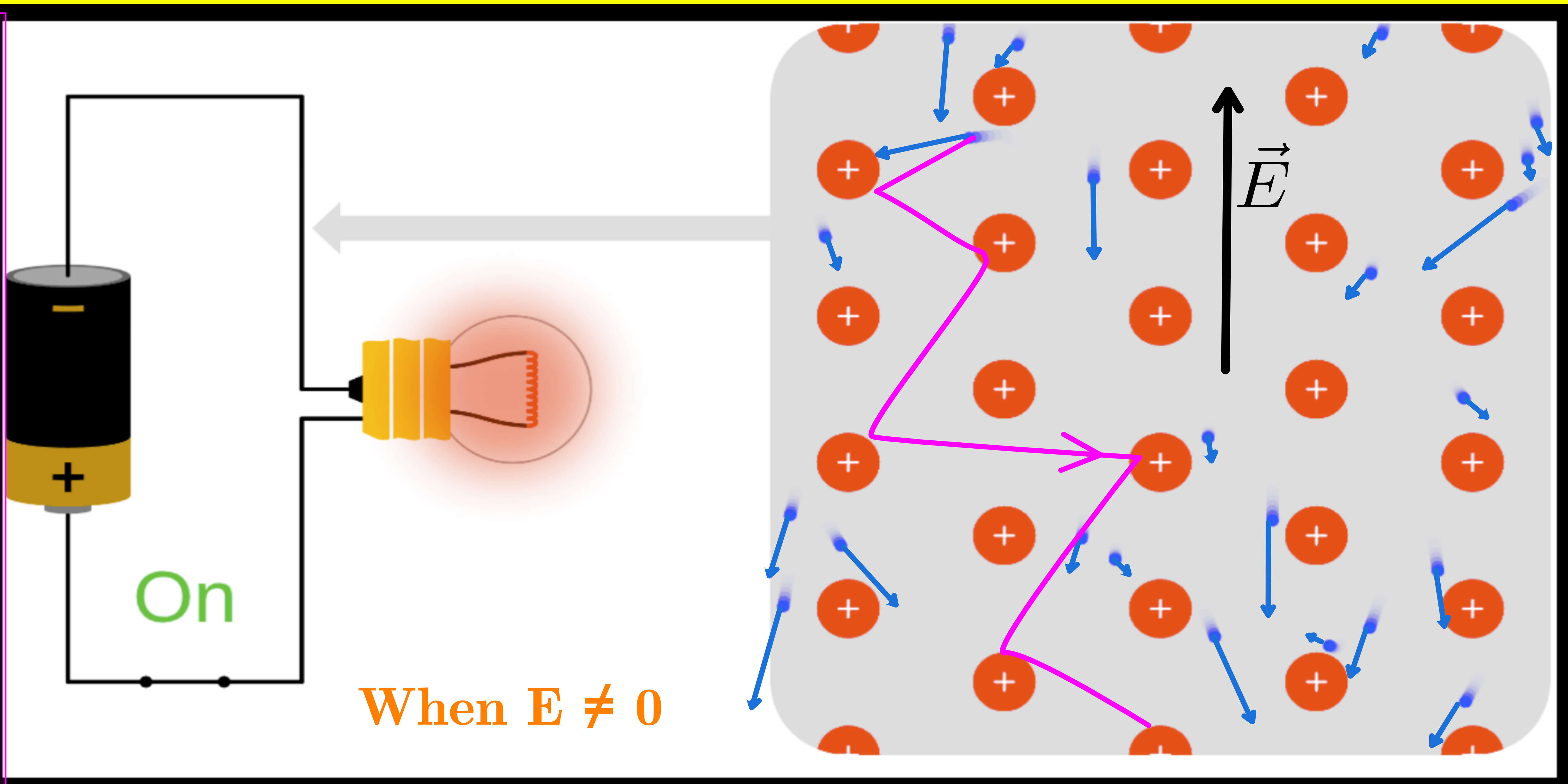

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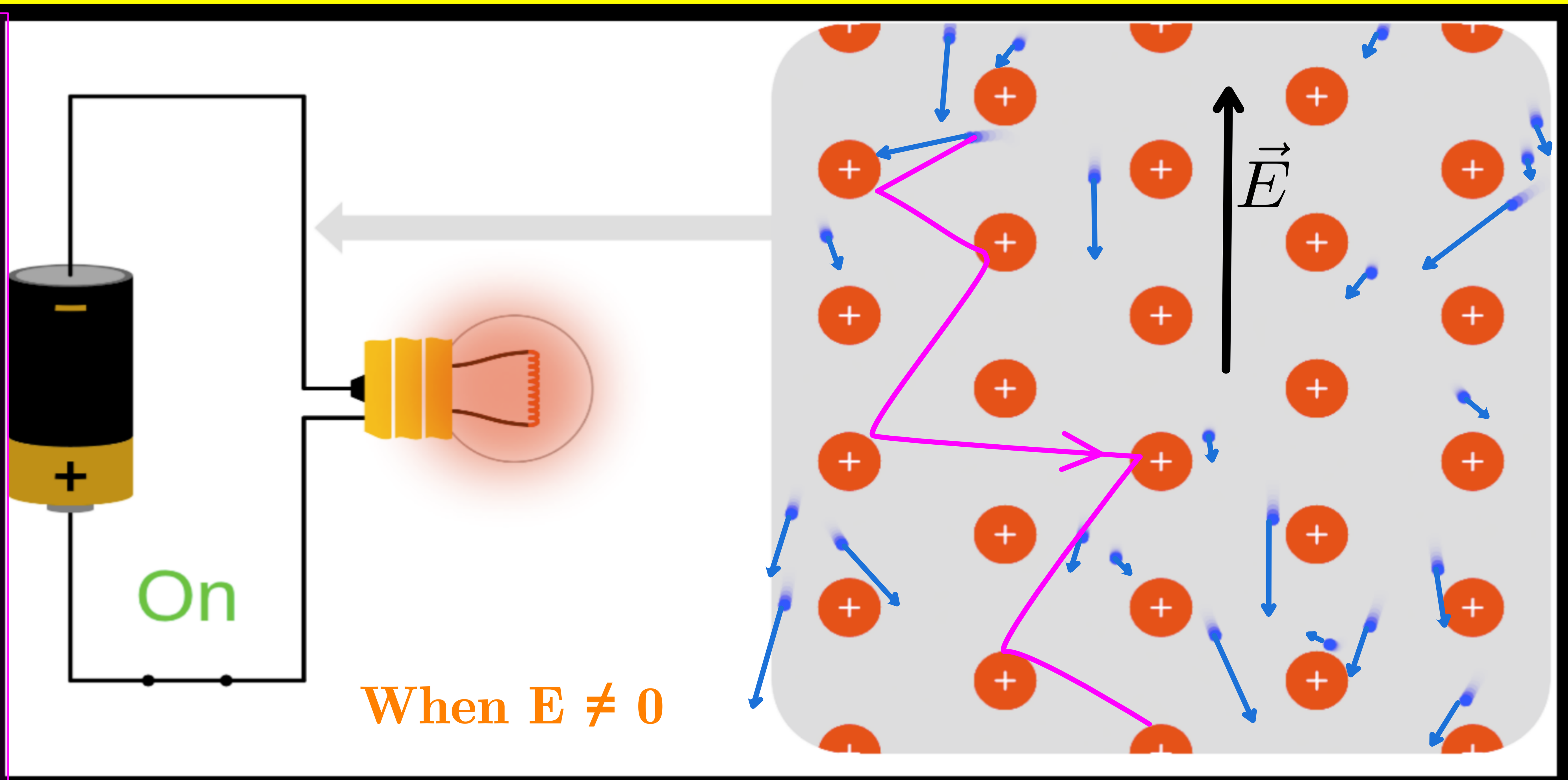

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* Drift velocity : The average velocity with which the free electrons get drifted in the opposite direction of the applied electric field.

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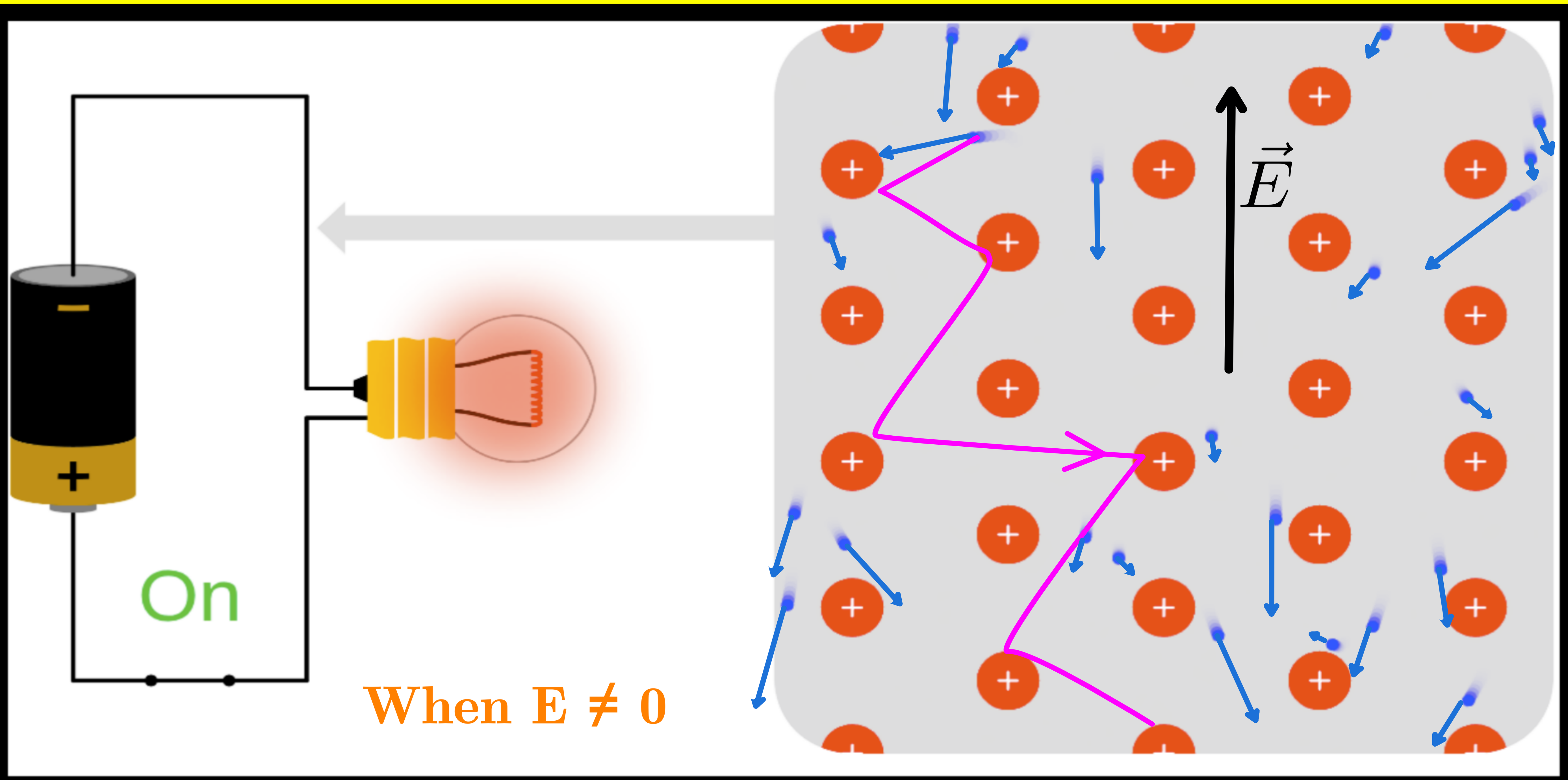

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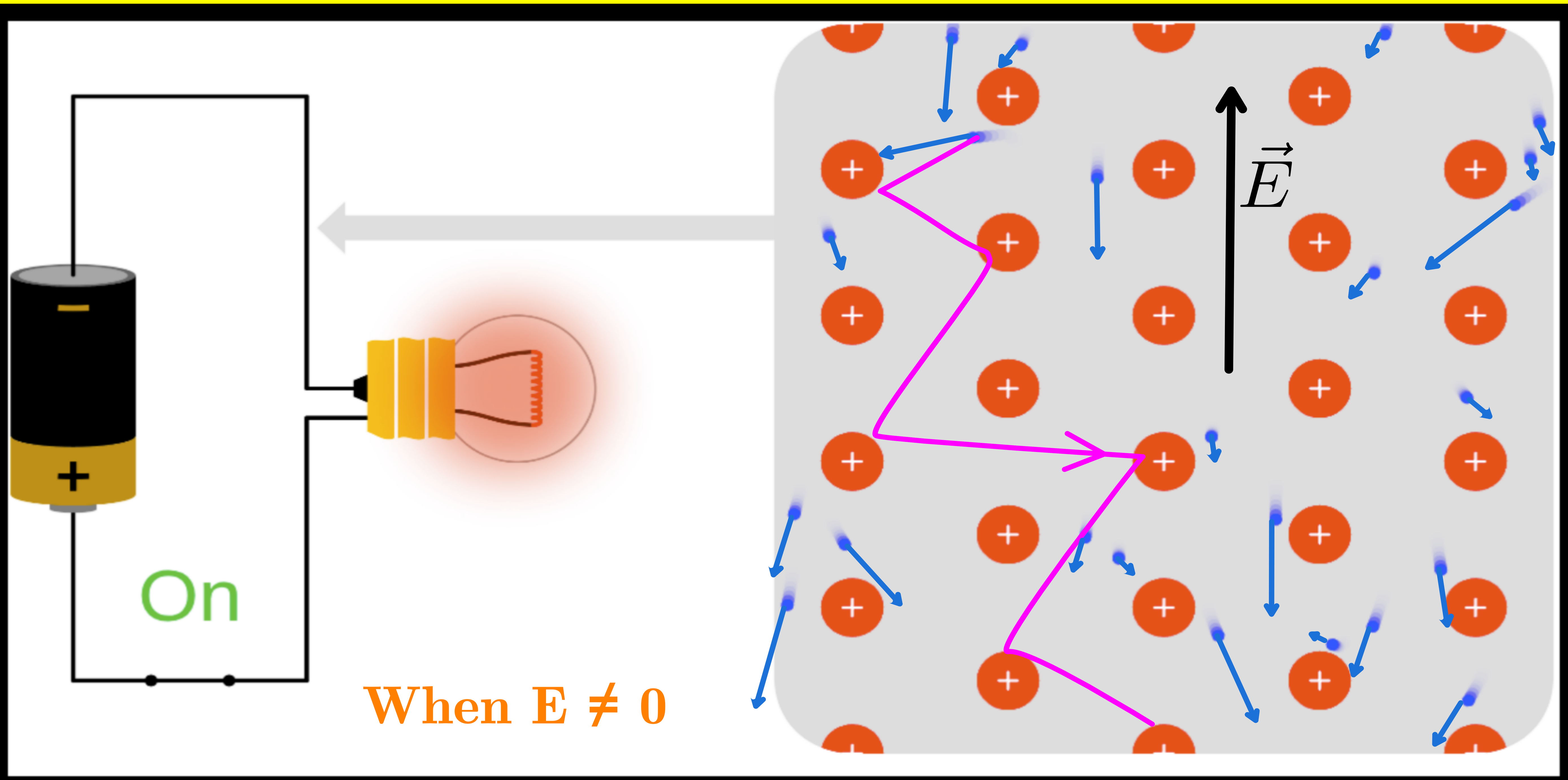

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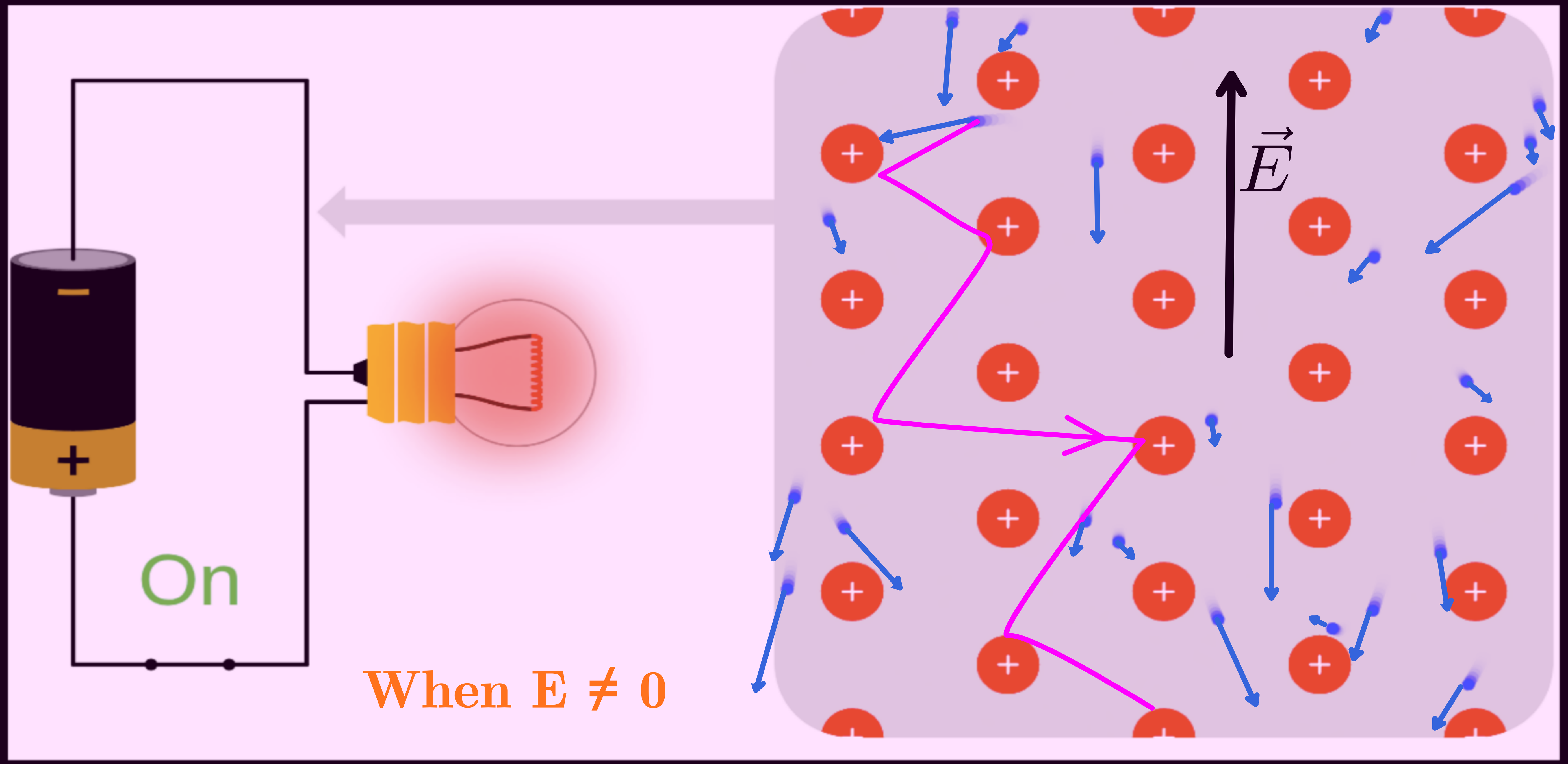

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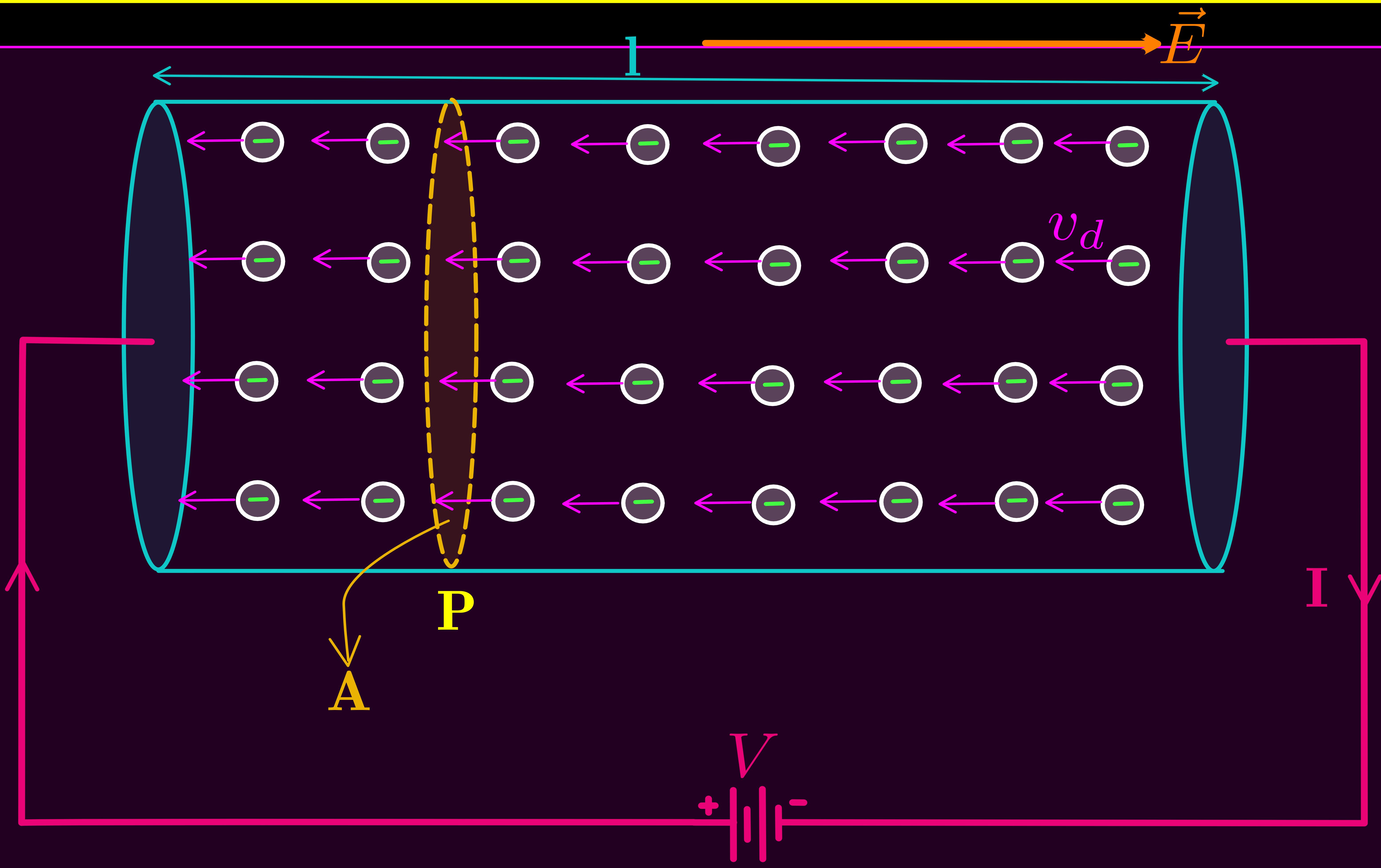

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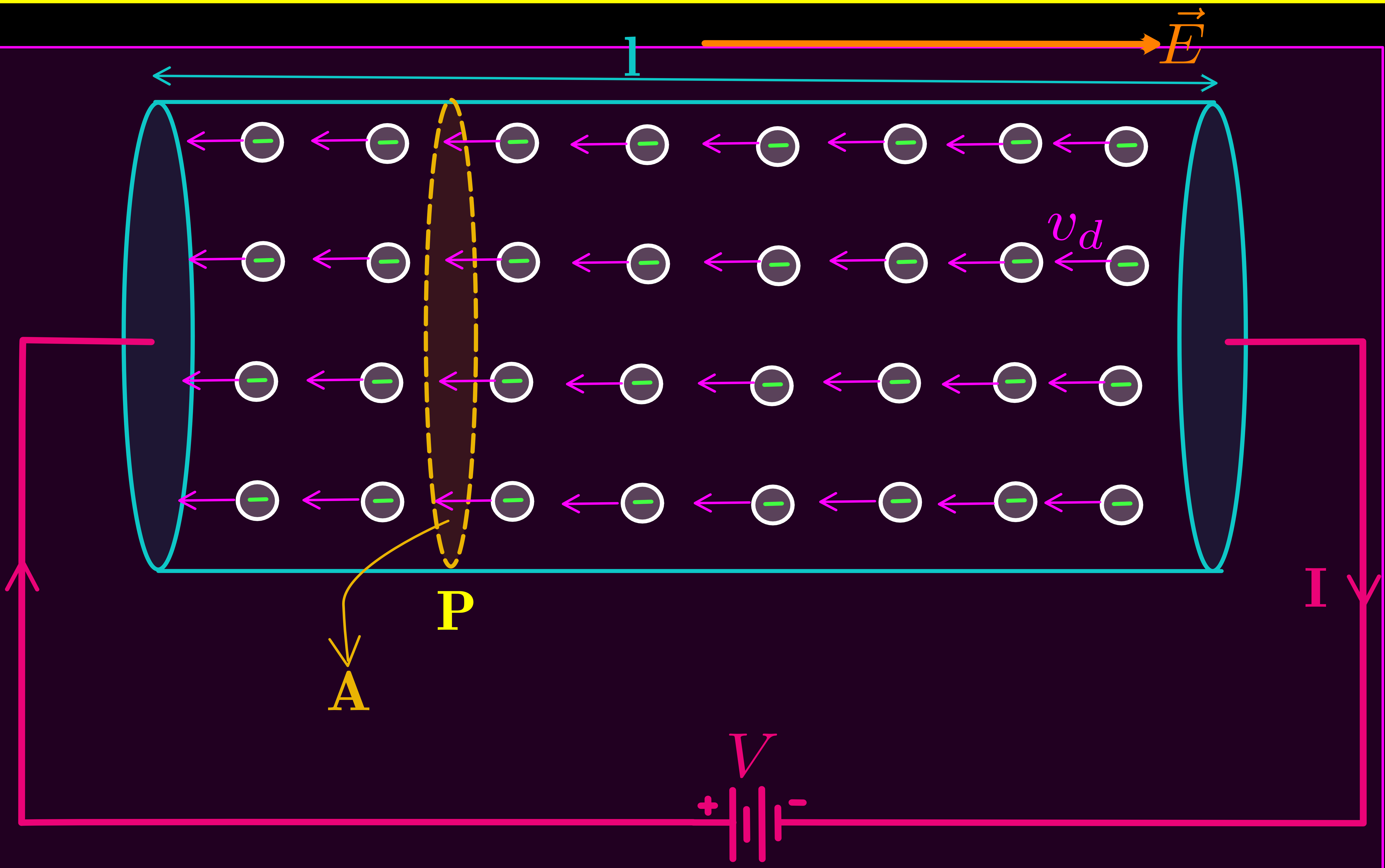

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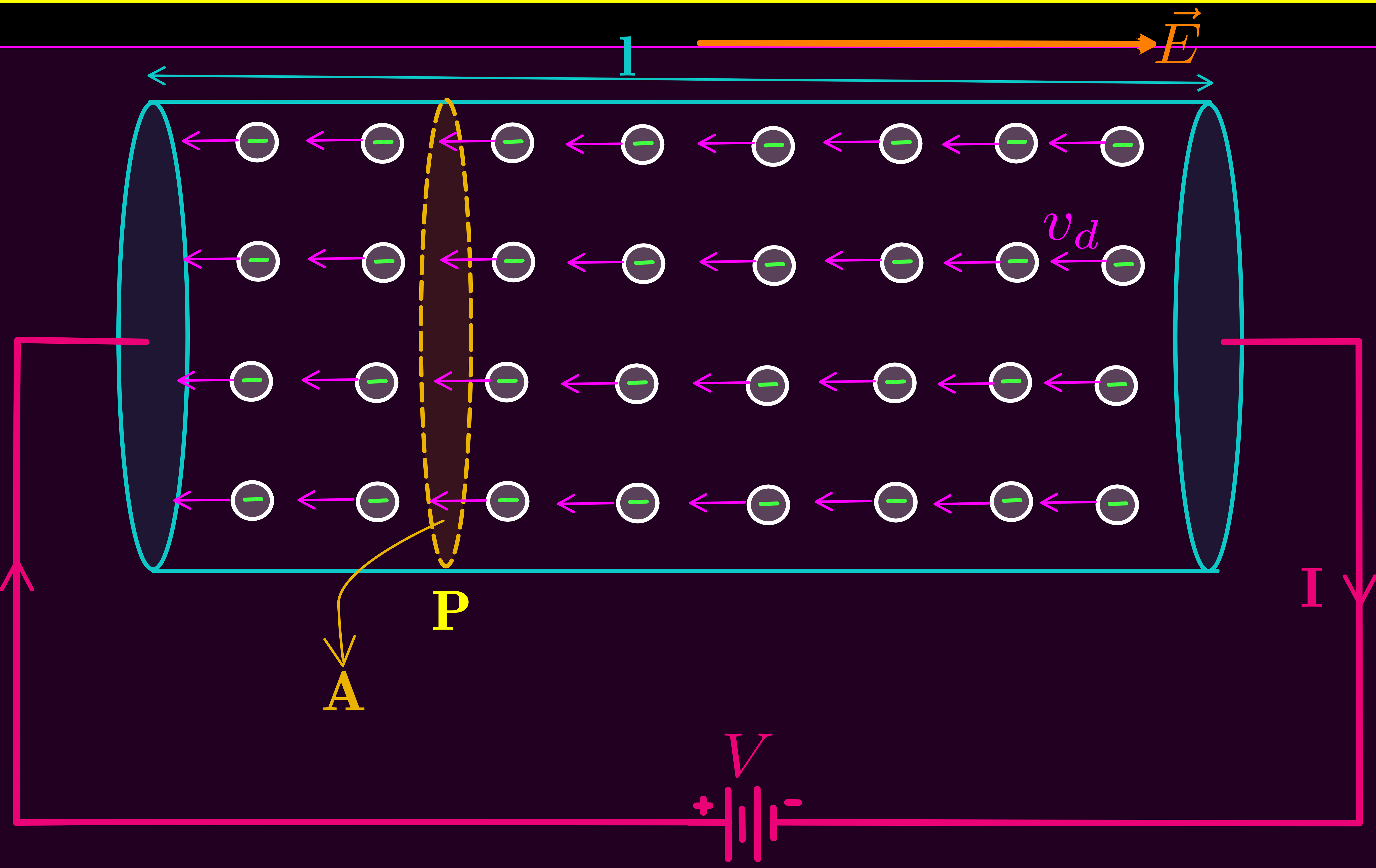

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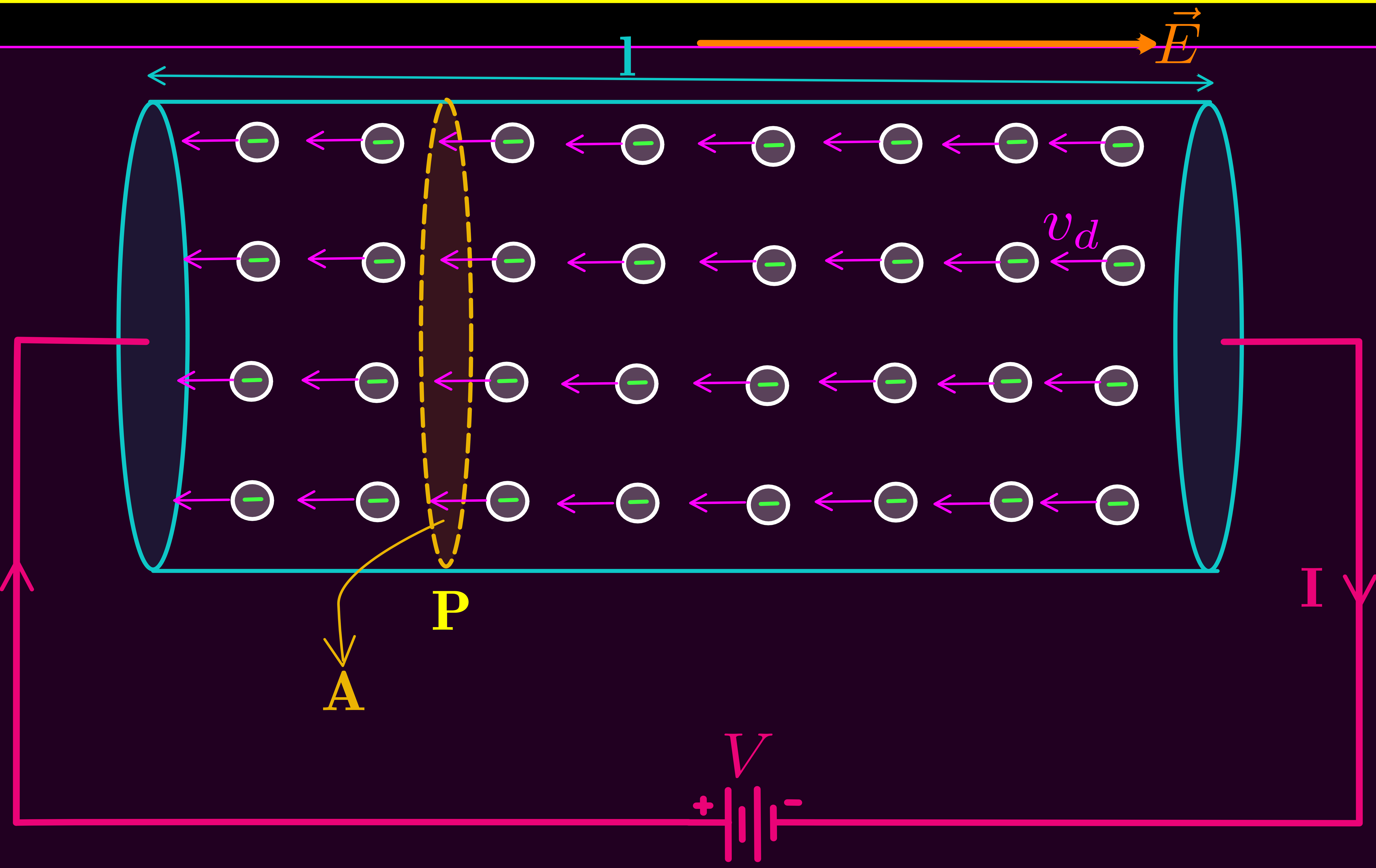

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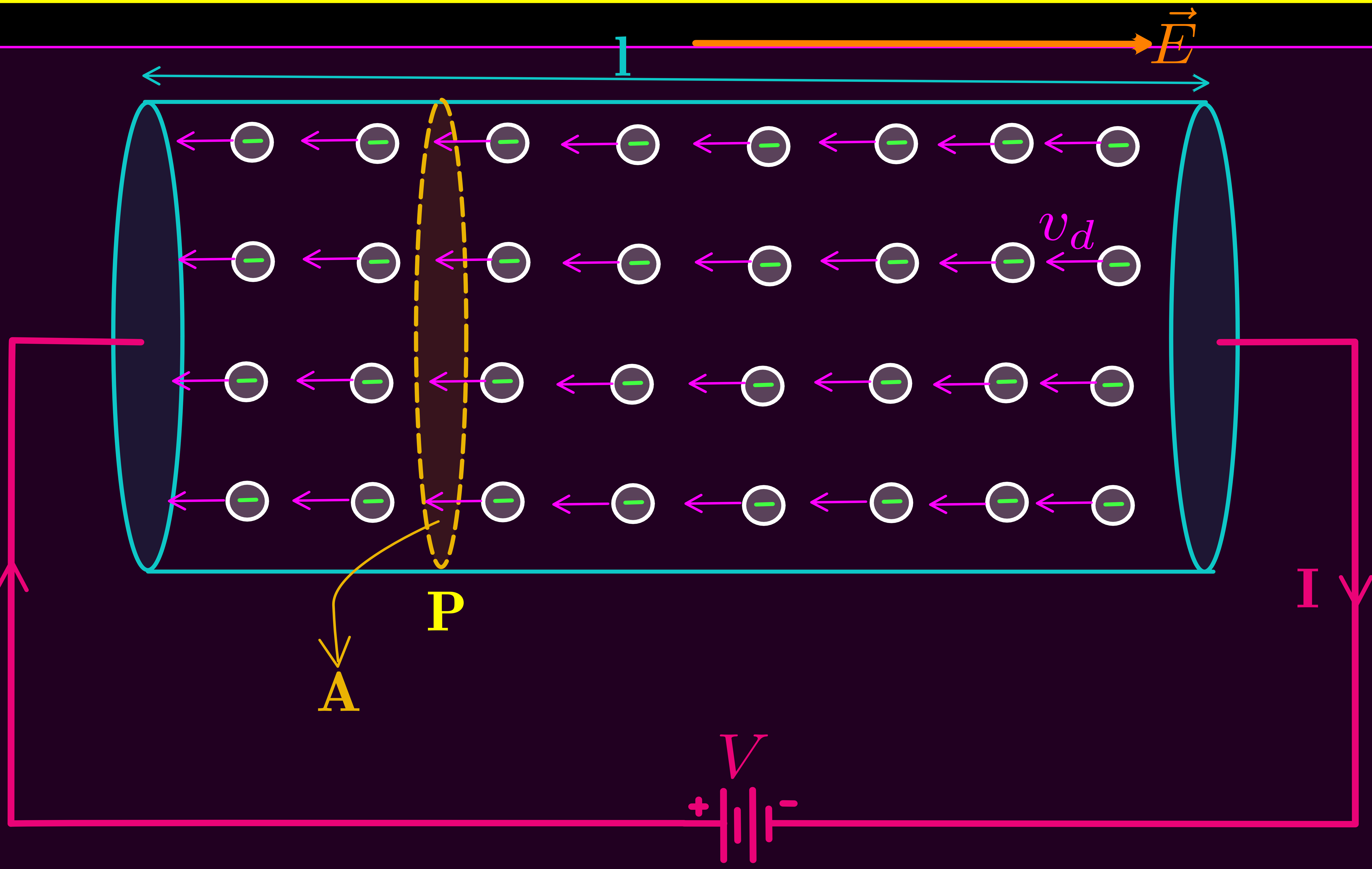

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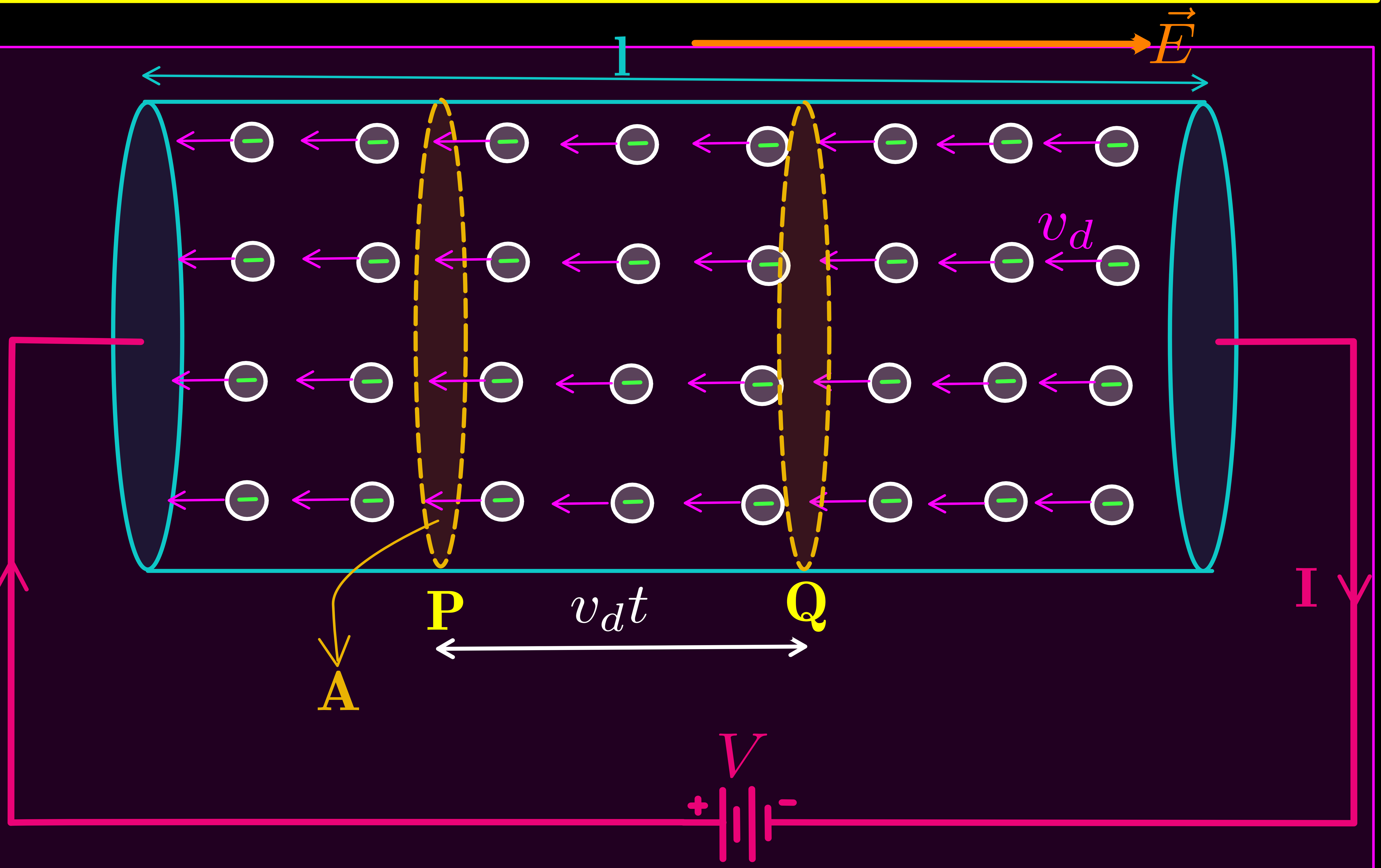

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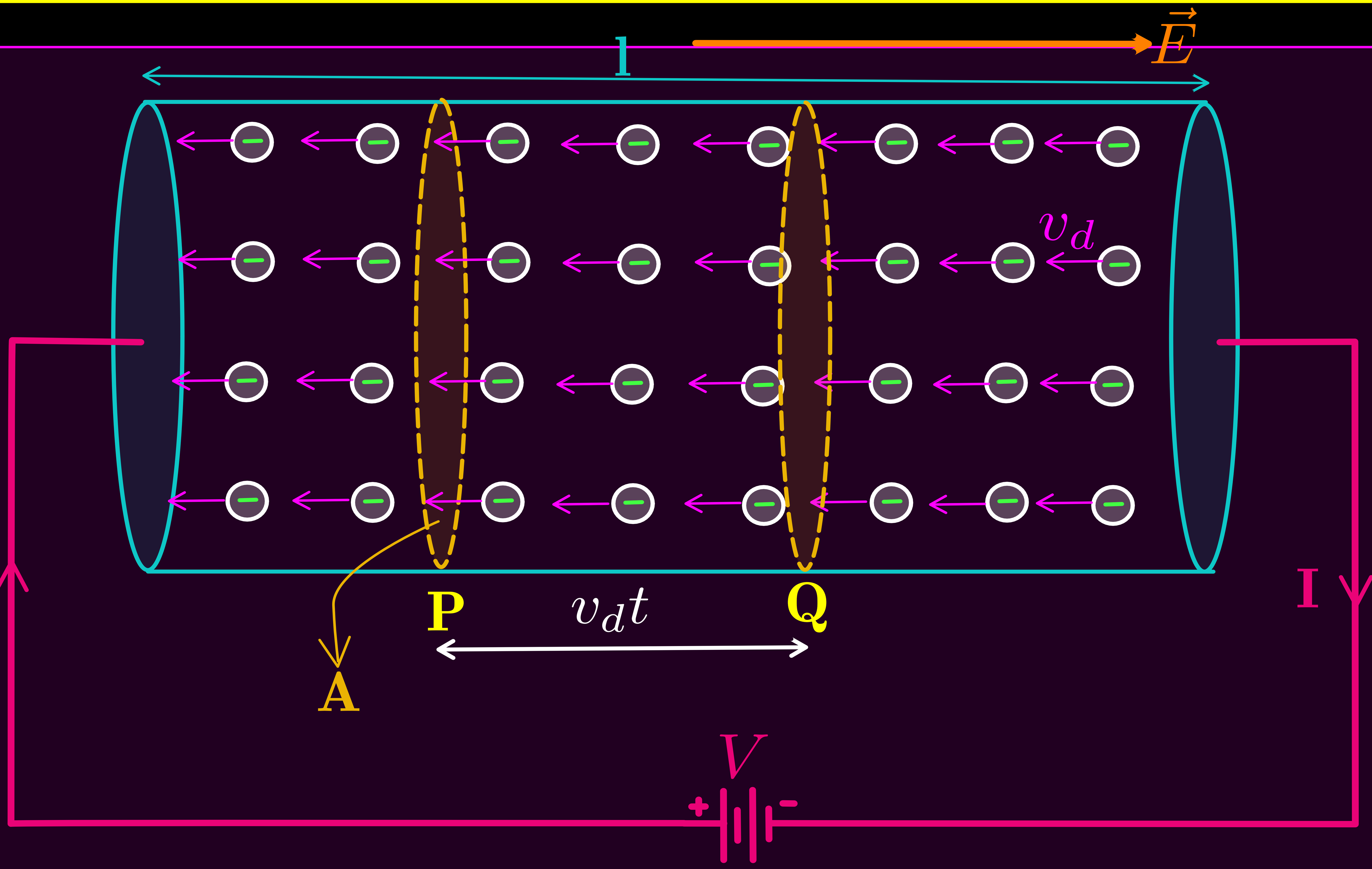

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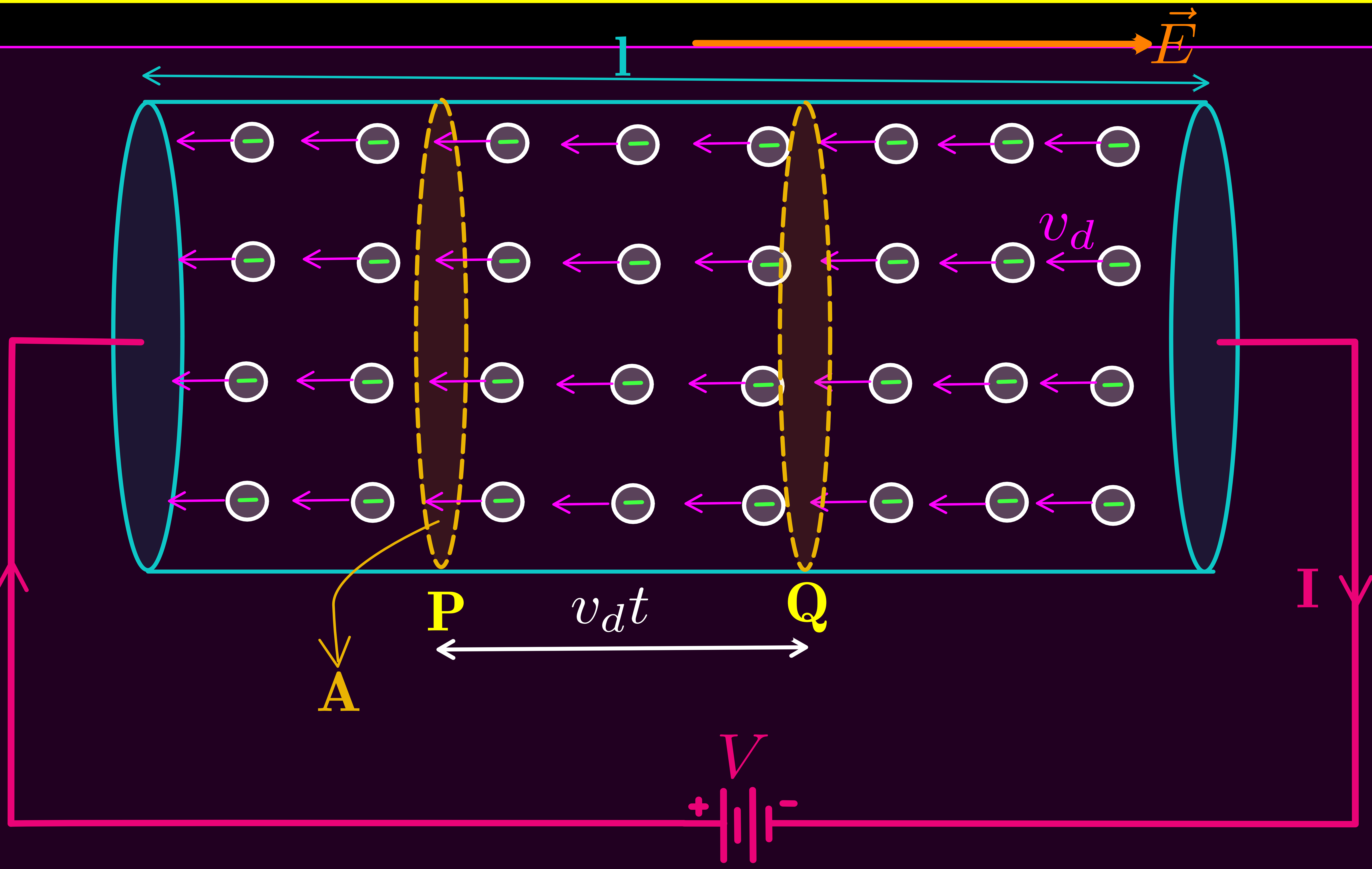

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$$= n \times A v_d \ t$$

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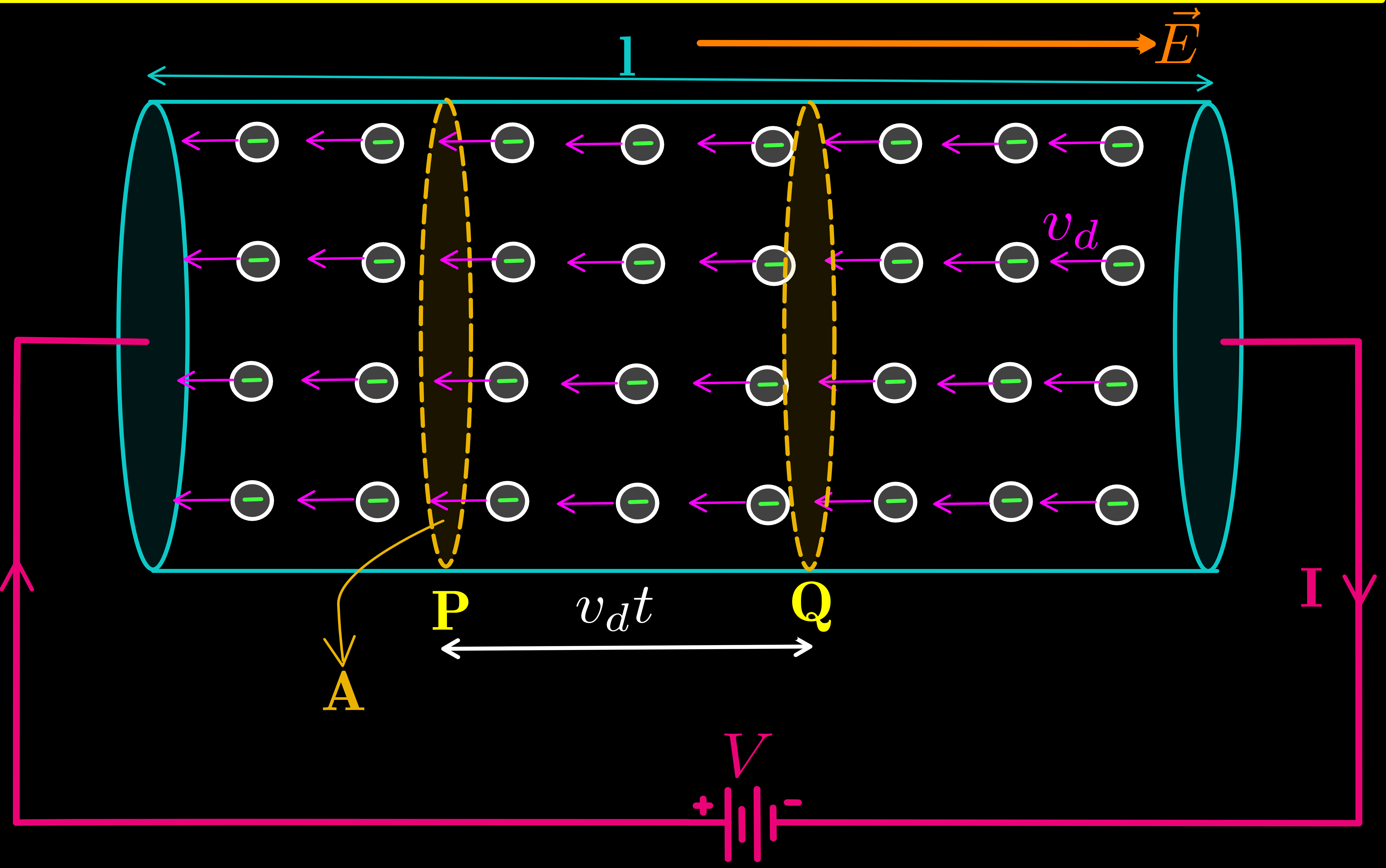

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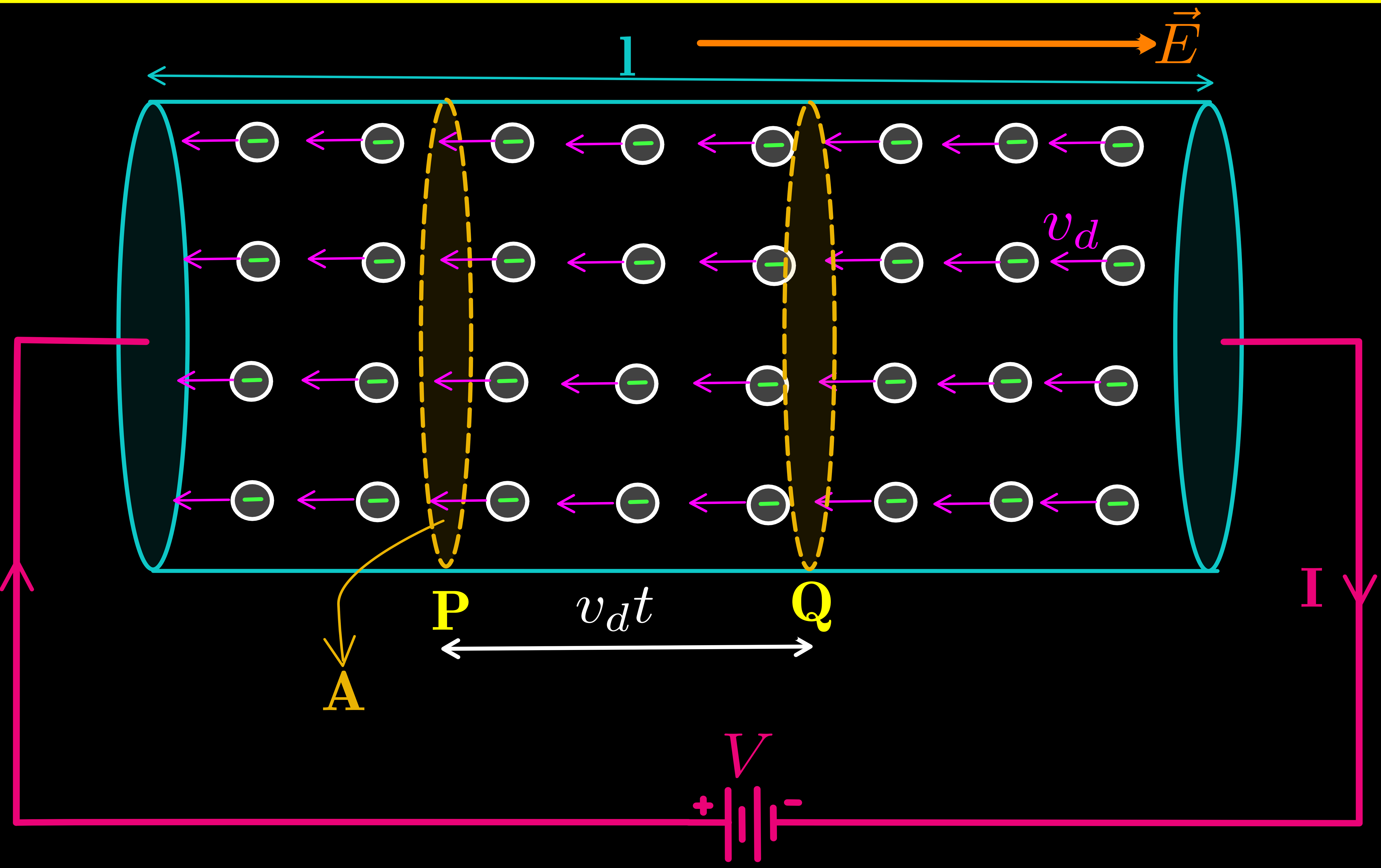

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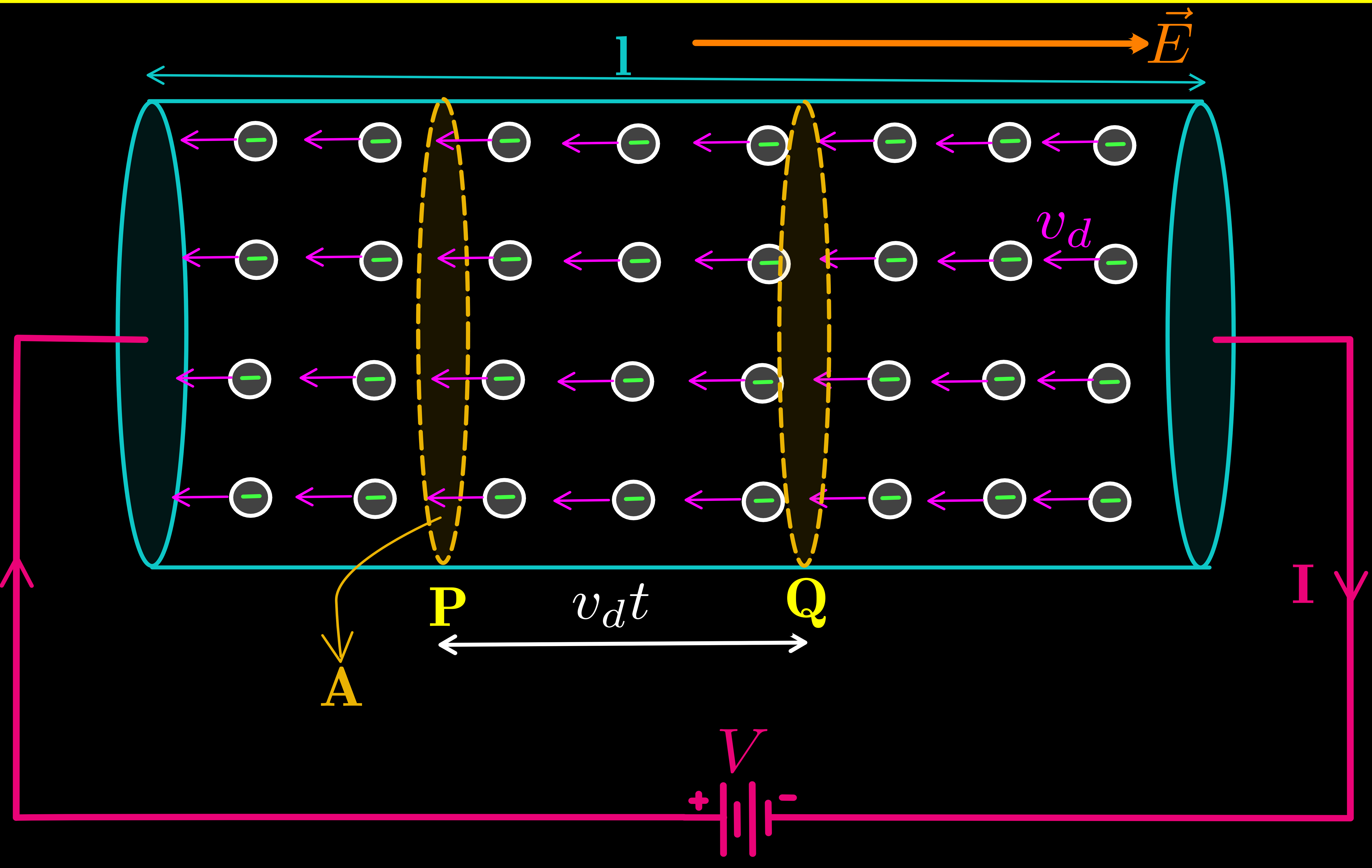

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and $\boxed{\text{Resistivity } \rho = \frac{1}{\sigma} = \frac{m_e}{ne^2\tau}}$

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Ex 9: There is a current of 40 A in a wire of 10^{-6} square metre area of cross-section, if the number of free electrons per cubic metre is 10^{29}, then the drift velcoity will be

$(a)\ 1.25 \times 10^{3}\ m/s$ $\qquad$ $(b)\ 2.5 \times 10^{-3}\ m/s$

$(c)\ 25 \times 10^{-3}\ m/s$ $\qquad$ $(d)\ 25 \times 10^{3}\ m/s$

Example 3.1 (a) Estimate the average drift speed of conduction electrons in a copper wire of cross-sectional area 1.0×10^{-7} m^2 carrying a current of 1.5 A. Assume that each copper atom contributes roughly one conduction electron. The density of copper is 9.0×10^3 kg/m^3, and its atomic mass is 63.5 u. (b) Compare the drift speed obtained above with, (i) thermal speeds of copper atoms at ordinary temperatures, (ii) speed of propagation of electric field along the conductor which causes the drift motion.

3. 13 The number density of free electrons in a copper conductor estimated in Example 3.1 is $8.5 \times 10^{28}\ m^{-3}$. How long does an electron take to drift from one end of a wire 3.0 m long to its other end? The area of cross-section of the wire is $2.0 \times 10^{-6}\ m^2$ and it is carrying a current of 3.0 A.

Example 3.2

(a) In Example 3.1, the electron drift speed is estimated to be only a few mm s^{-1} for currents in the range of a few amperes? How then is current established almost the instant a circuit is closed?

(b) The electron drift arises due to the force experienced by electrons in the electric field inside the conductor. But force should cause acceleration. Why then do the electrons acquire a steady average drift speed?

(c) If the electron drift speed is so small, and the electron's charge is small, how can we still obtain large amounts of current in a conductor?

(d) When electrons drift in a metal from lower to higher potential, does it mean that all the 'free' electrons of the metal are moving in the same direction?

(e) Are the paths of electrons straight lines between successive collisions (with the positive ions of the metal) in the (i) absence of electric field, (ii) presence of electric field?

Solution

(a) Electric field is established throughout the circuit, almost instantly (with the speed of light) causing at every point a *local electron drift.* Establishment of a current does not have to wait for electrons from one end of the conductor travelling to the other end. However, it does take a little while for the current to reach its steady value.

(b) Each 'free' electron does accelerate, increasing its drift speed until it collides with a positive ion of the metal. It loses its drift speed after collision but starts to accelerate and increases its drift speed again only to suffer a collision again and so on. On the average, therefore, electrons acquire only a drift speed.

(c) Simple, because the electron number density is enormous, $\sim 10^{29}\ m^{-3}$.

(d) By no means. The drift velocity is superposed over the large random velocities of electrons.

(e) In the absence of electric field, the paths are straight lines; in the presence of electric field, the paths are, in general, curved.